

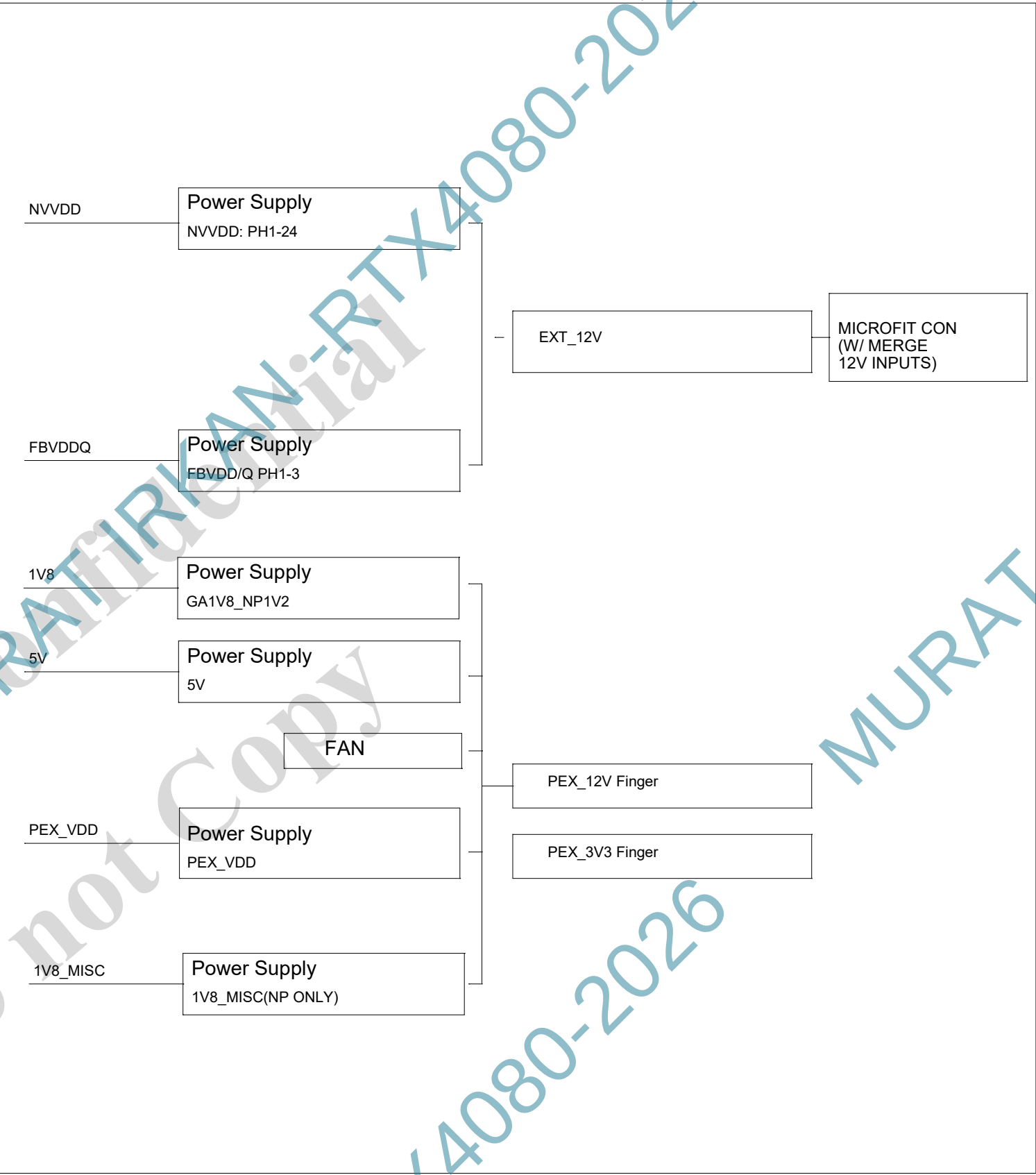
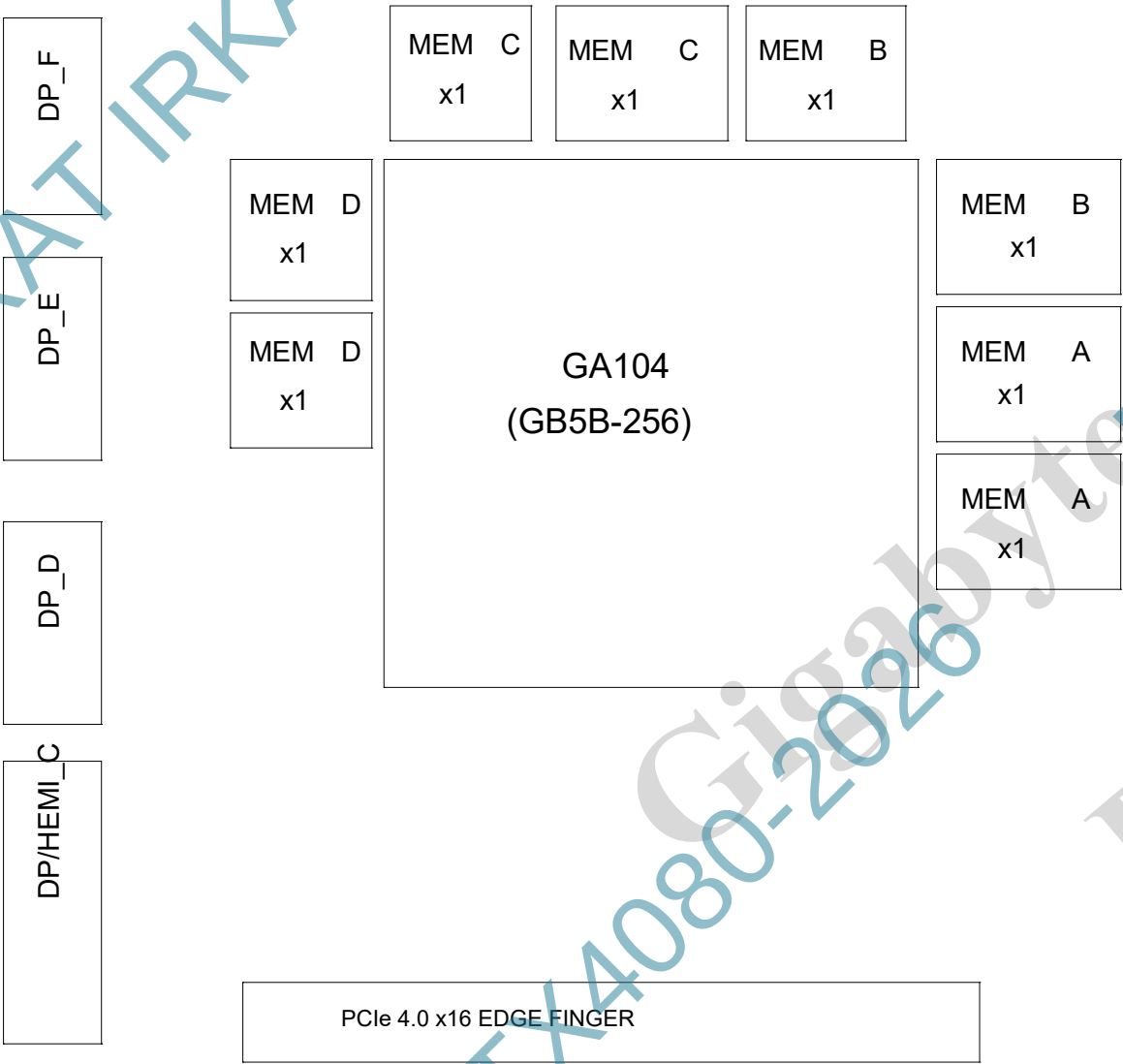
GA104/NP, GeForce SKU P-BOARD

400W-500W, 256bit, GDDR6x x16 16GB
DP + DP + DP + HDMI/DP

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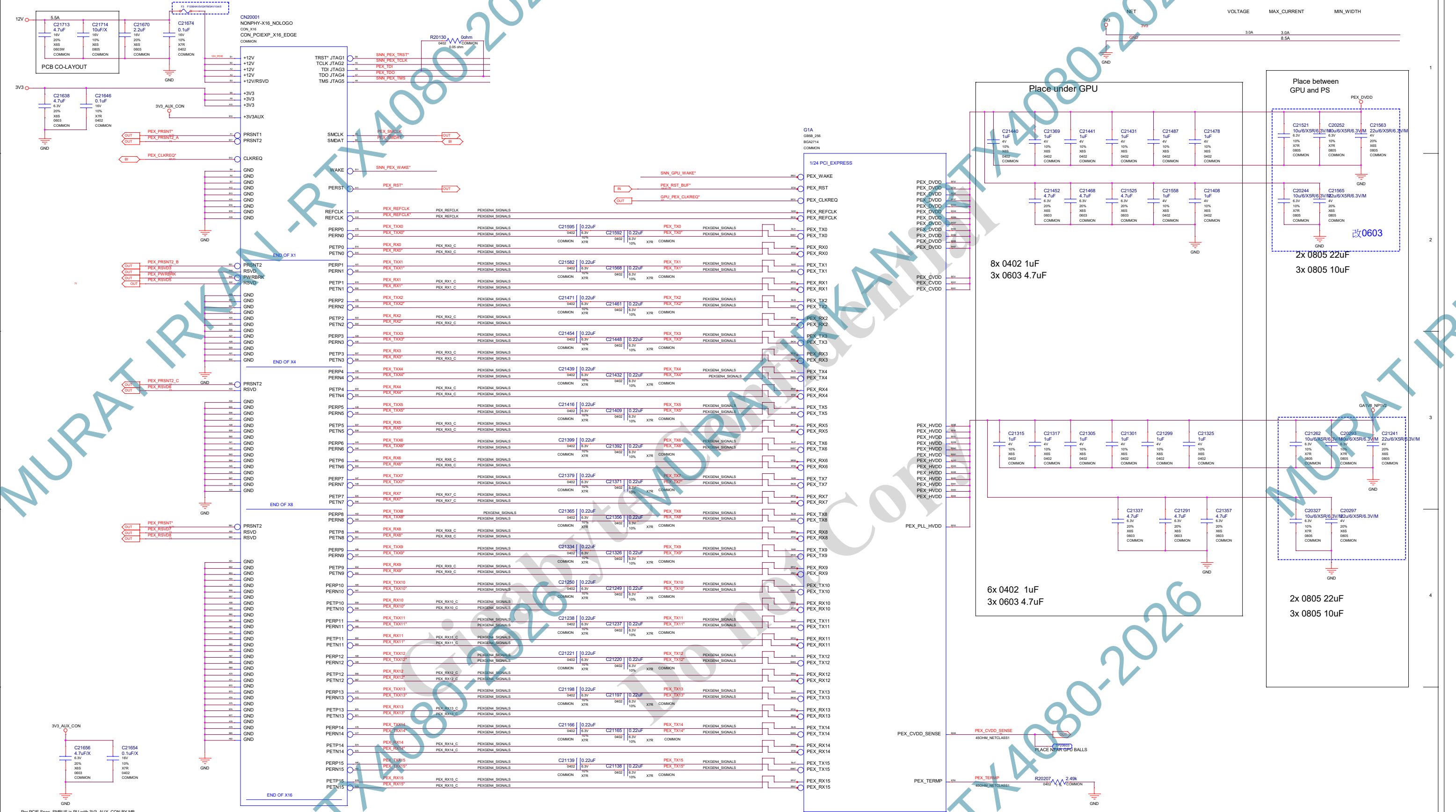


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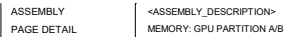
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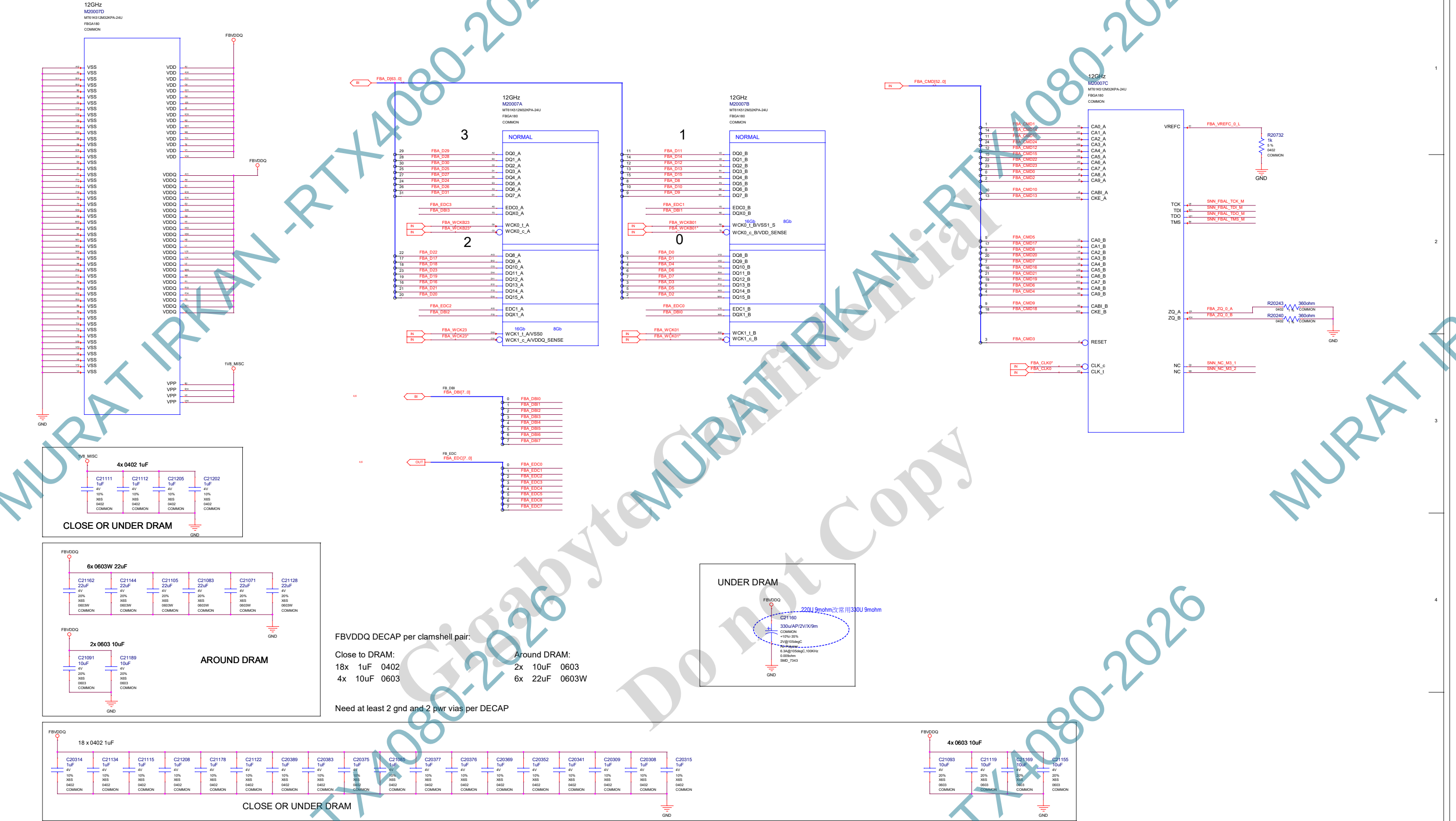
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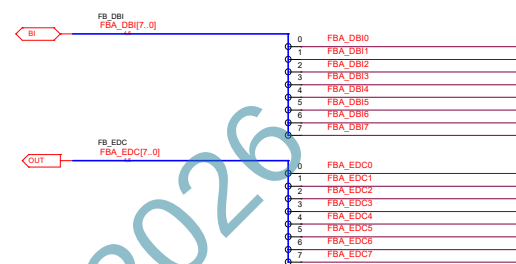
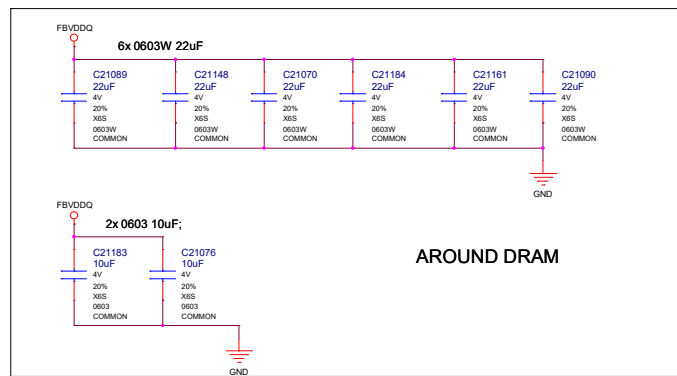
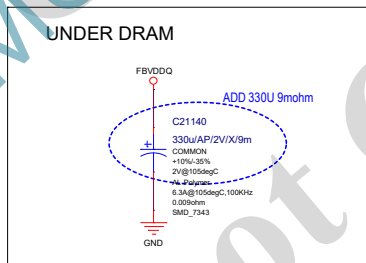
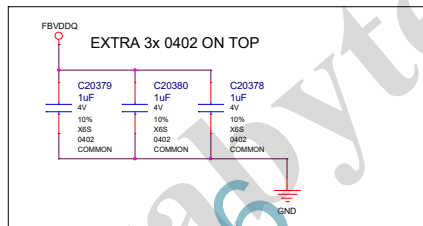
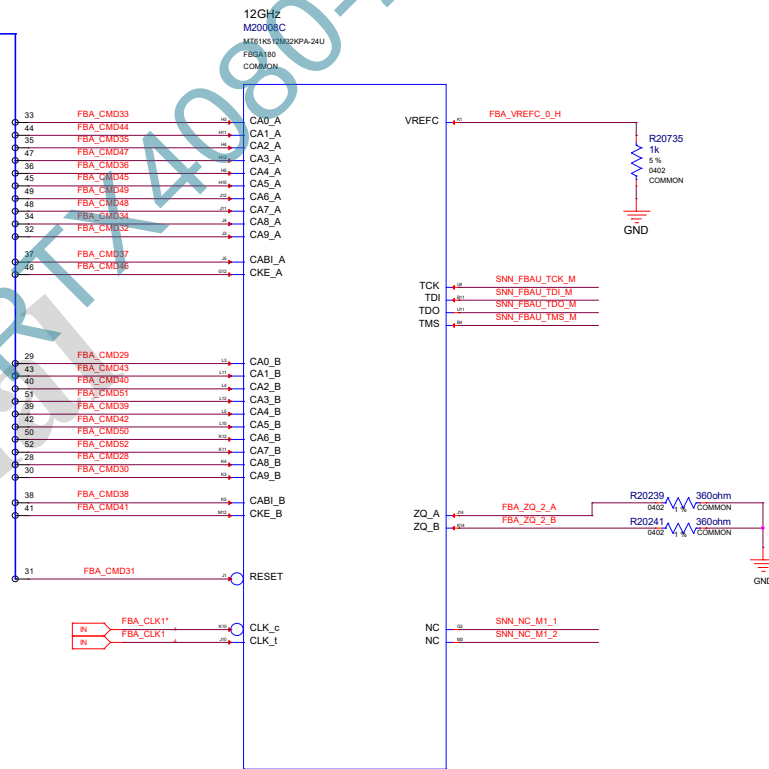
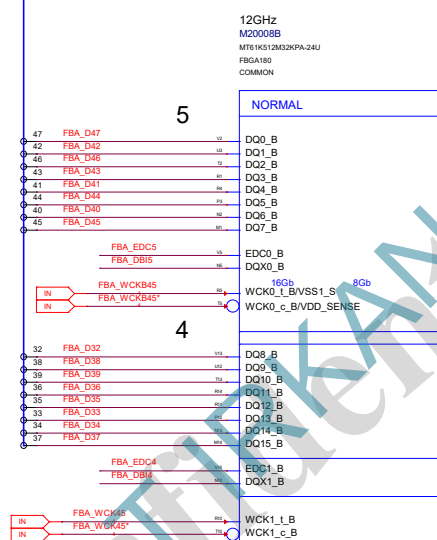
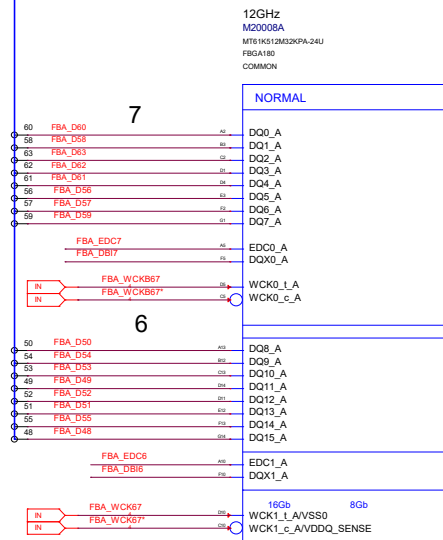


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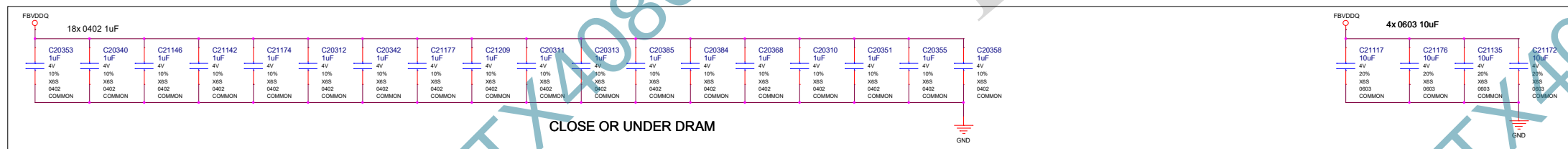
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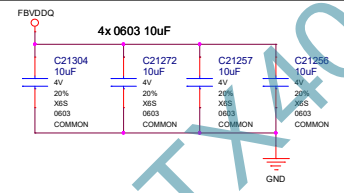
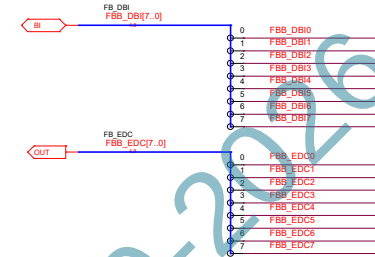
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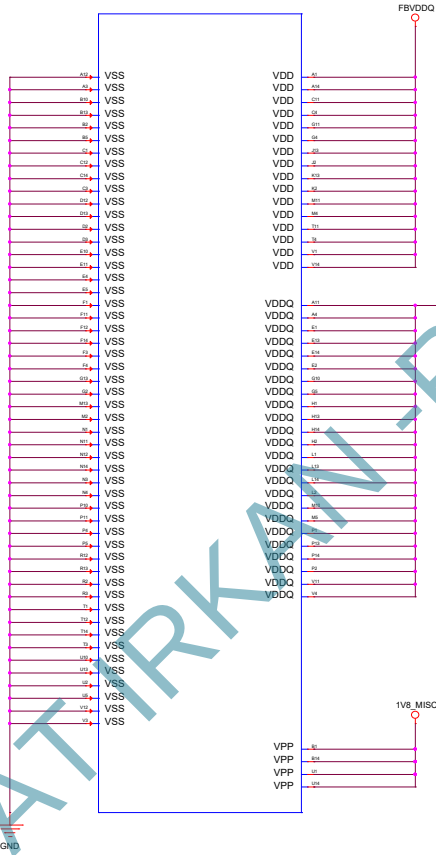
Need at least 2 gnd and 2 pwr vias per DECAP





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MT61K512M32KPA-24U
FBGA180
COMMON



FBVDDQ

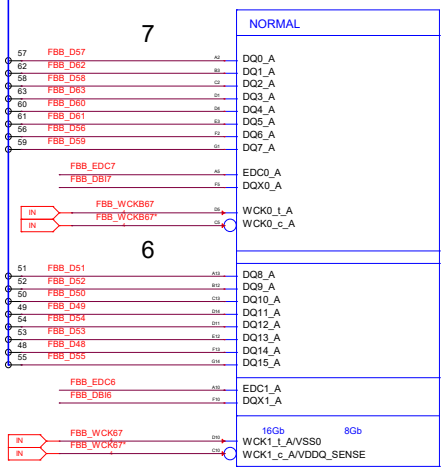
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1V8 MISC

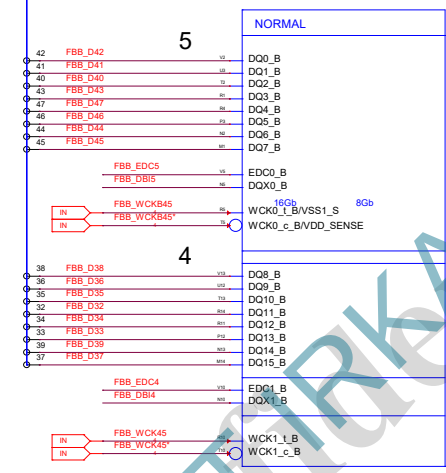
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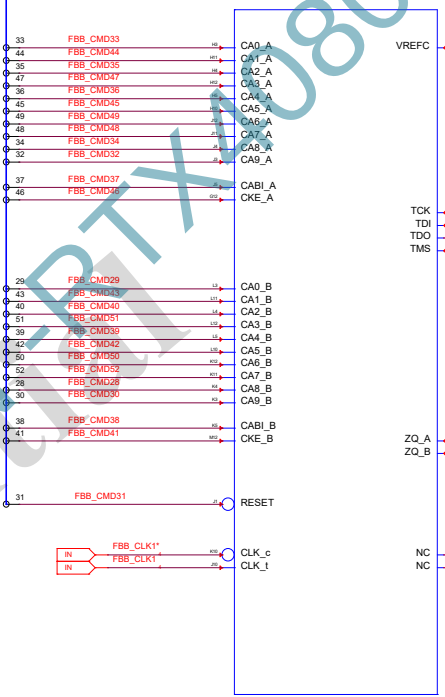
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12GHz
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12GHz
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VREFC

TCK

TDI

TDO

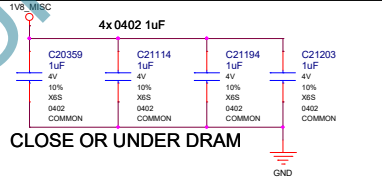
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ZQ_A

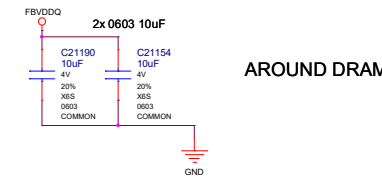
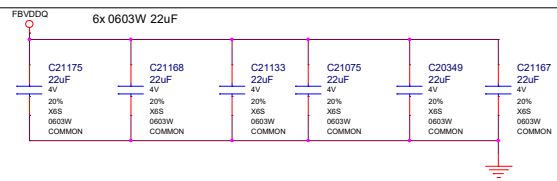
ZQ_B

NC

NC



CLOSE OR UNDER DRAM



AROUND DRAM

FBVDDQ DECAP per clamshell pair:

Close to DRAM:

18x 1uF 0402

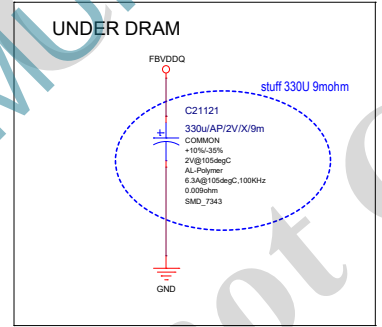
4x 10uF 0603

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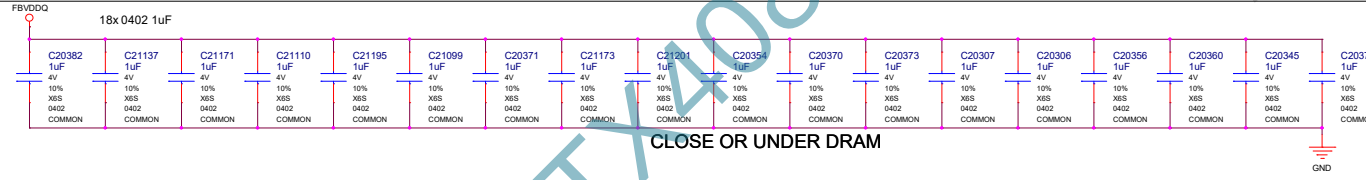
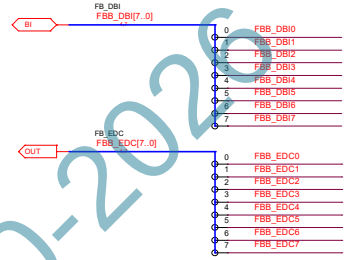
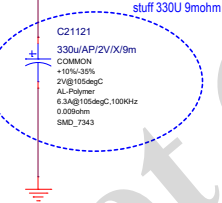
2x 10uF 0603

6x 22uF 0603W

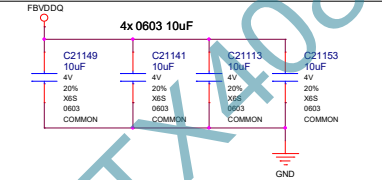
Need at least 2 gnd and 2 pwr vias per DECAP



UNDER DRAM



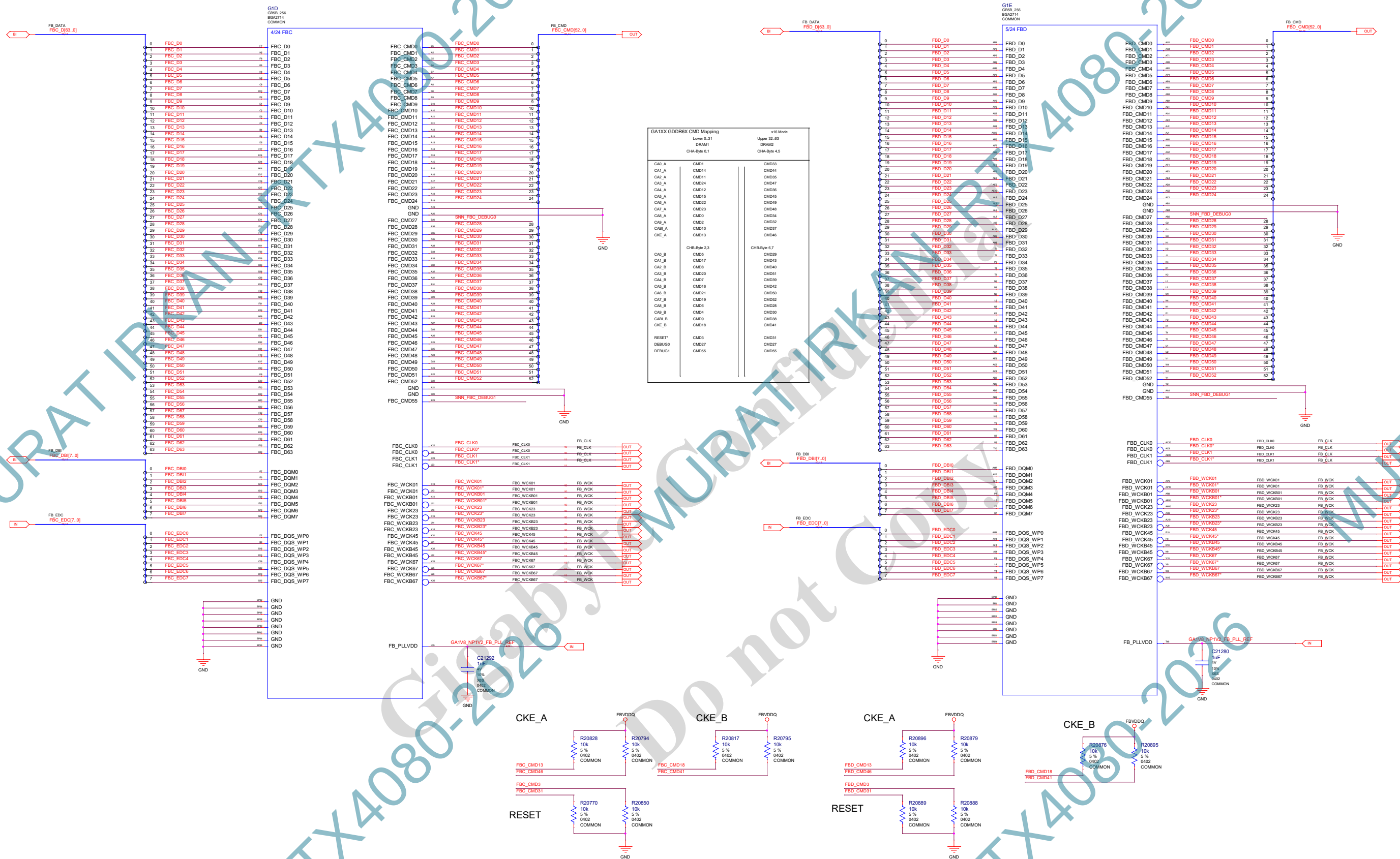
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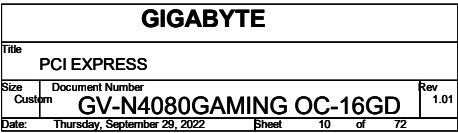
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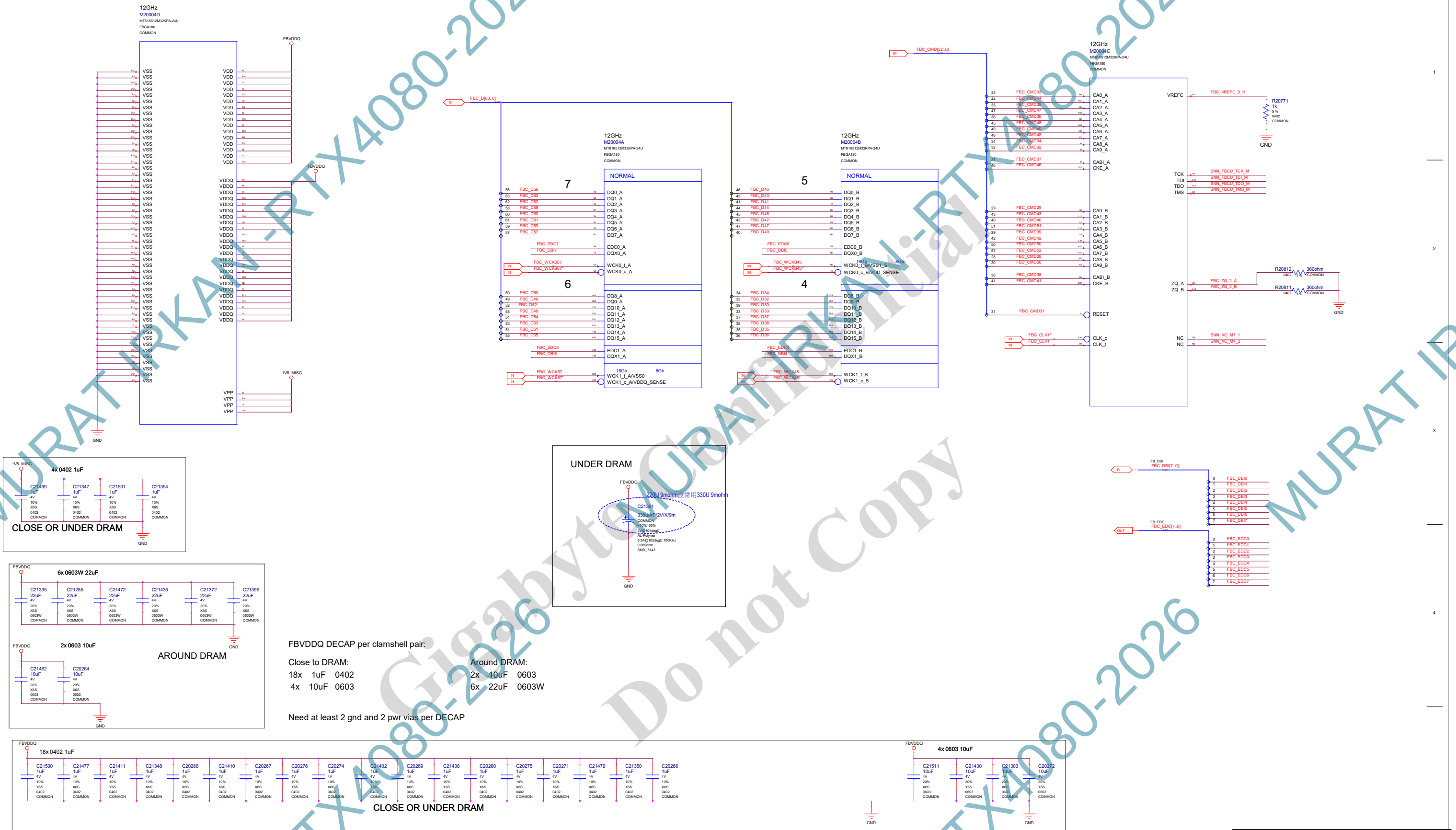


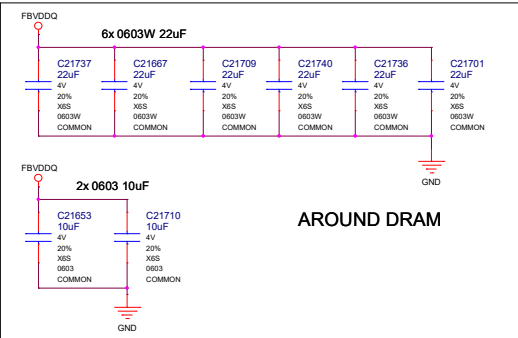
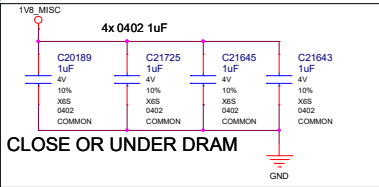
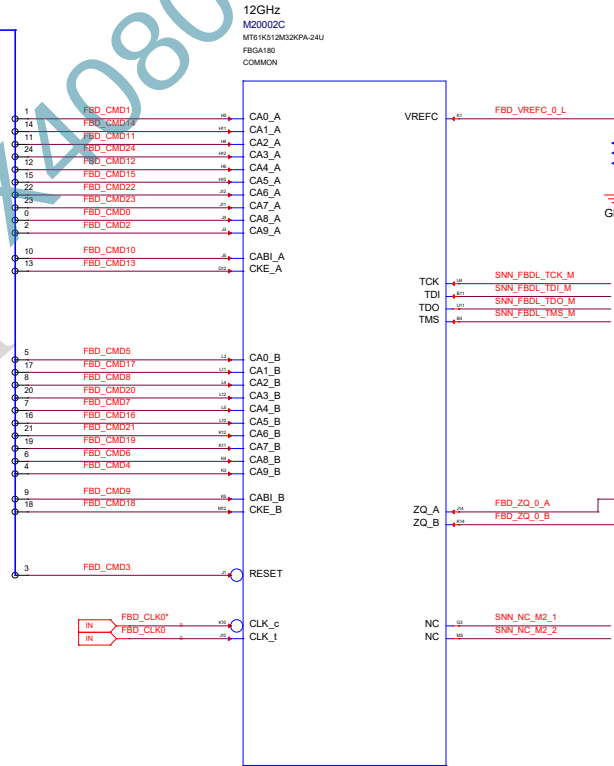
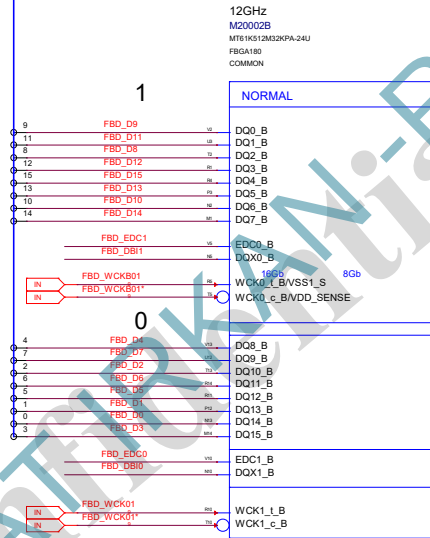
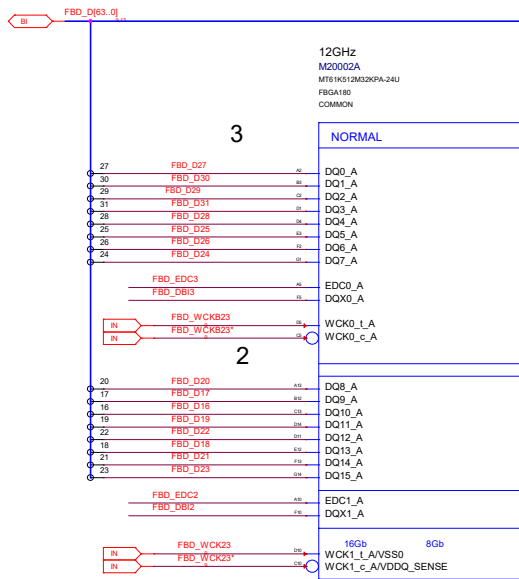
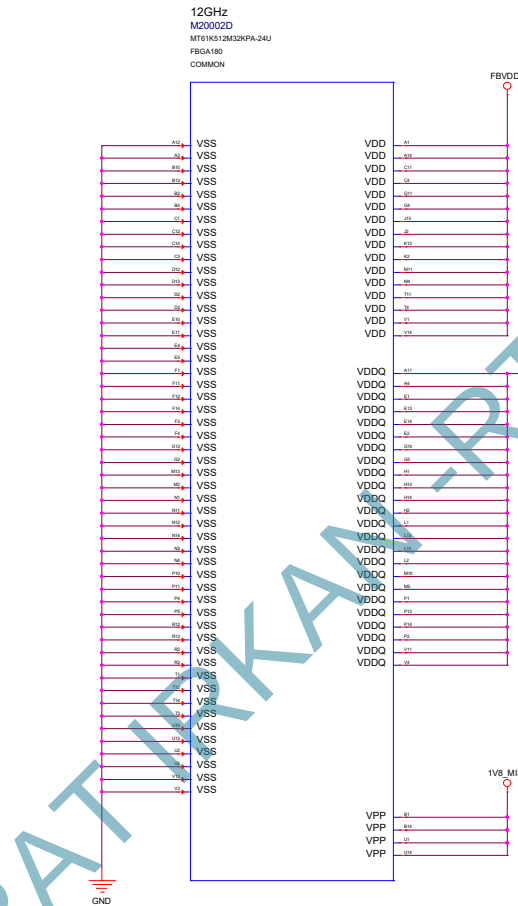
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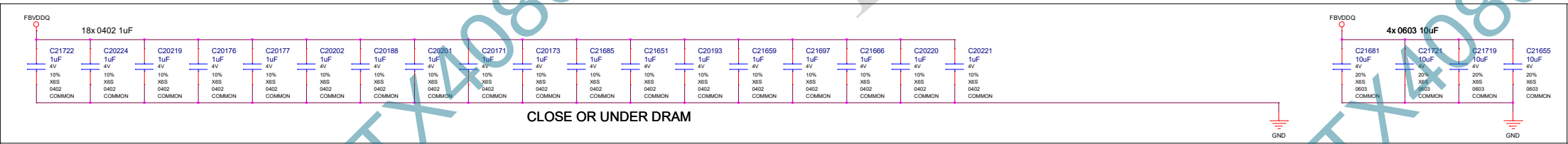
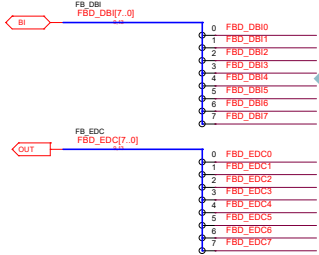
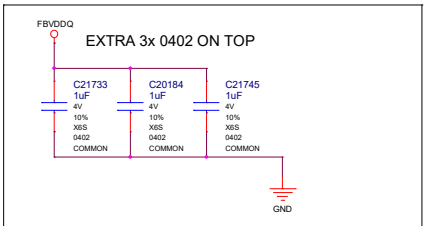
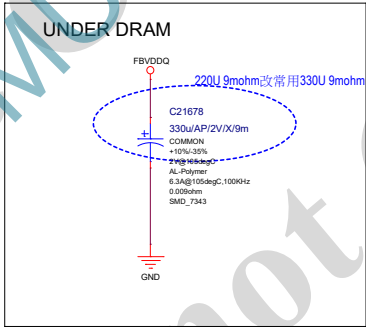


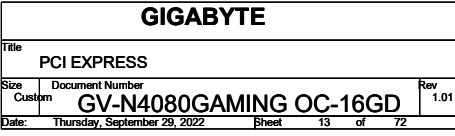


FBVDDQ DECAP per clamshell pair:

Close to DRAM: 18x 1uF 0402
4x 10uF 0603
Around DRAM: 2x 10uF 0603
6x 22uF 0603W

Need at least 2 gnd and 2 pwr vias per DECAP





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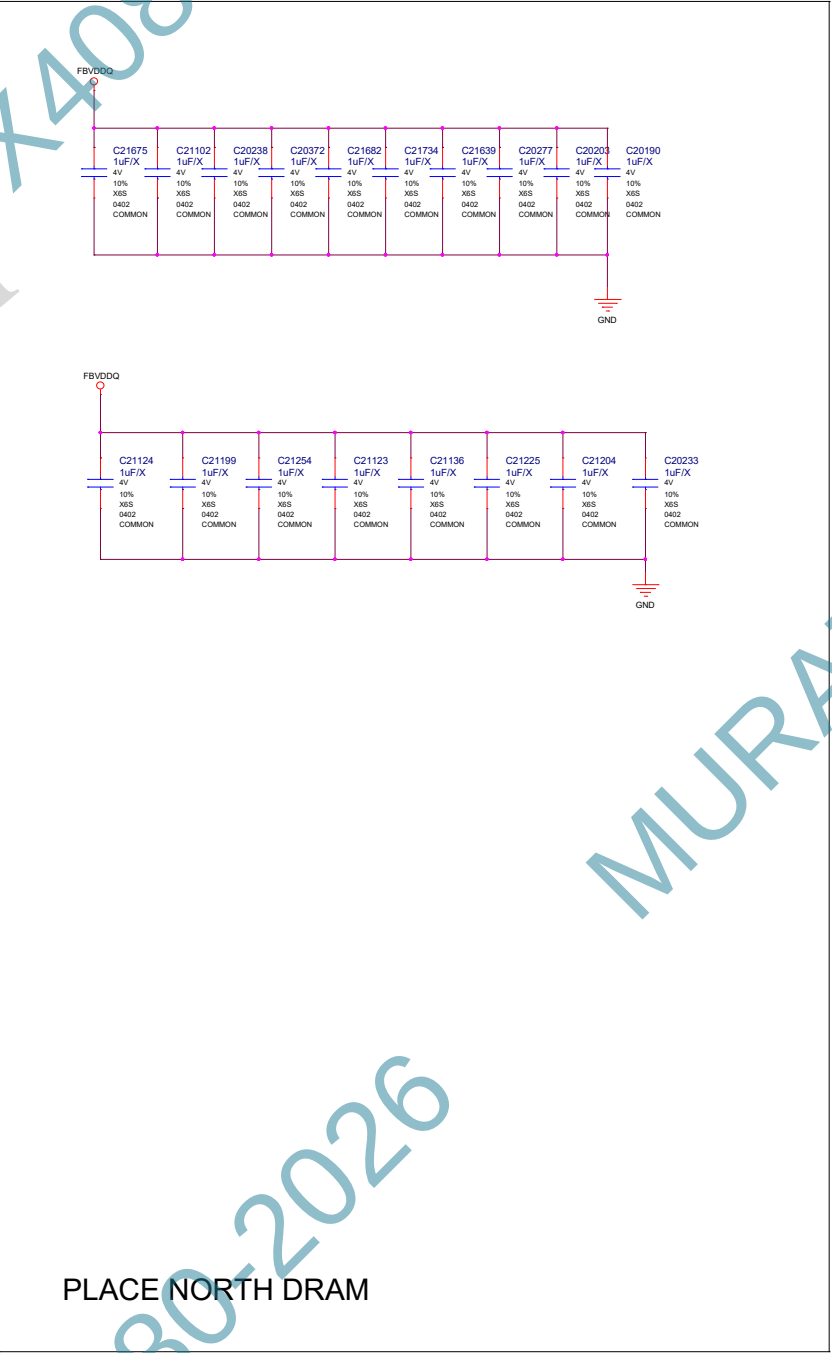
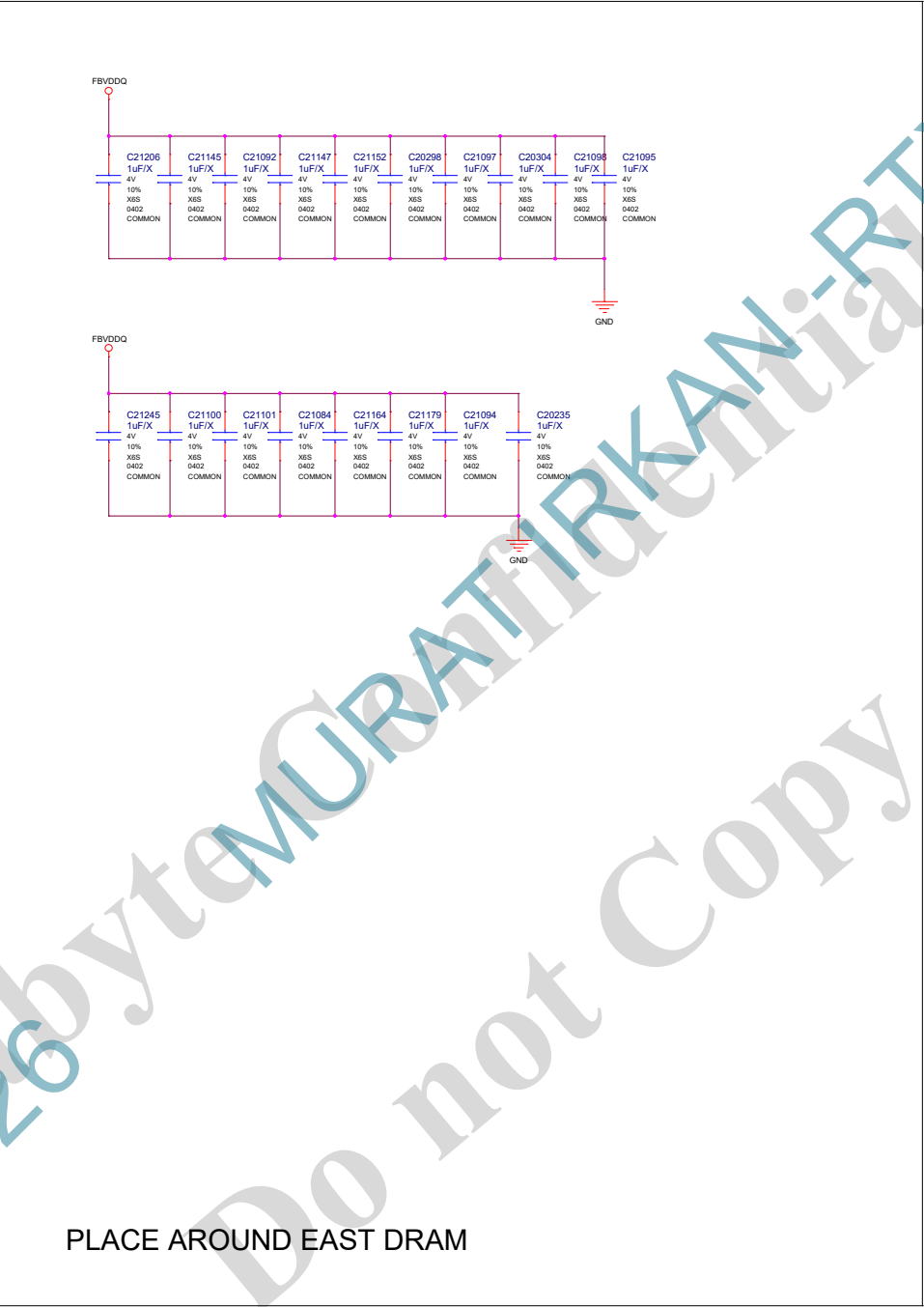
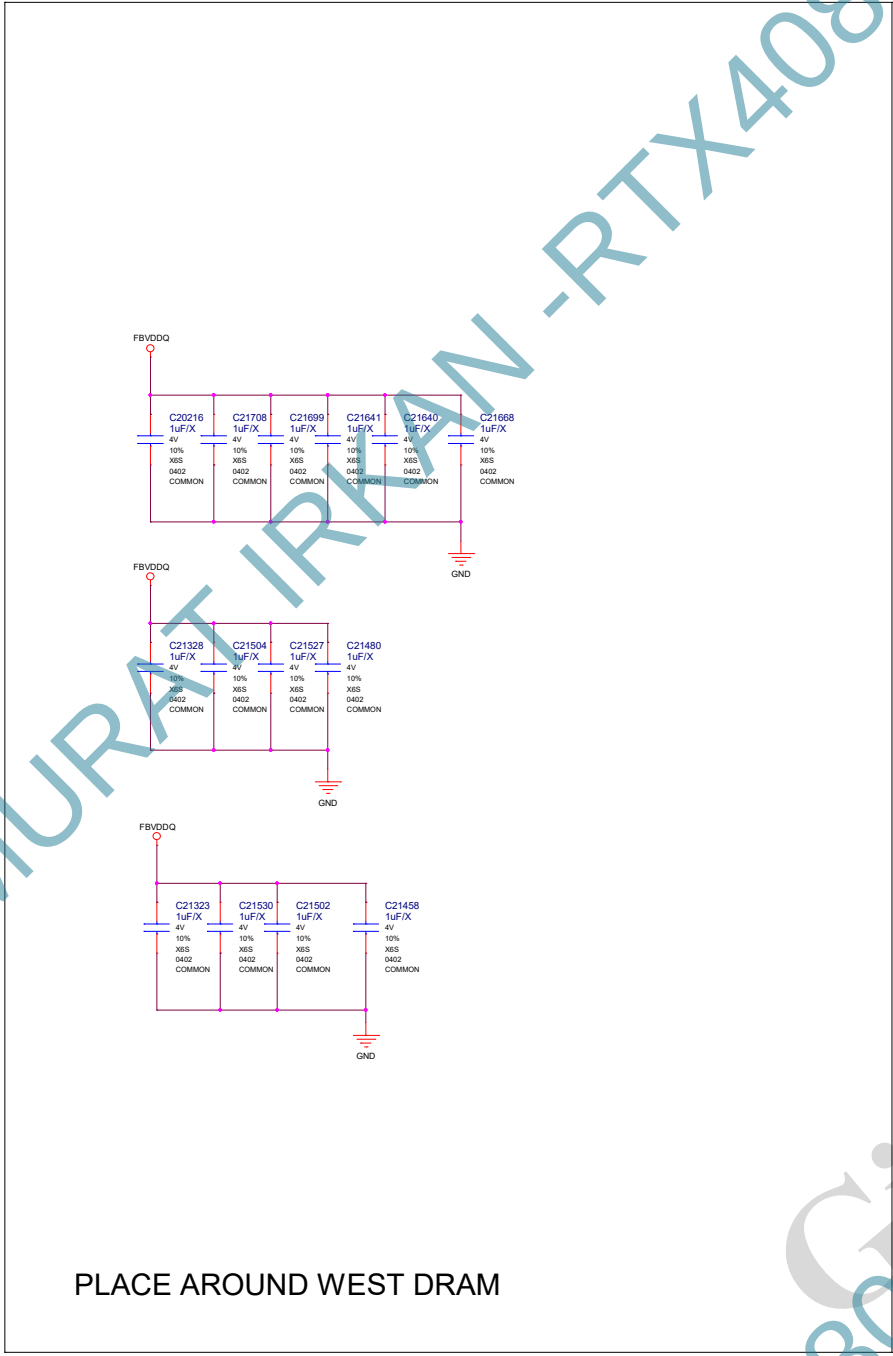
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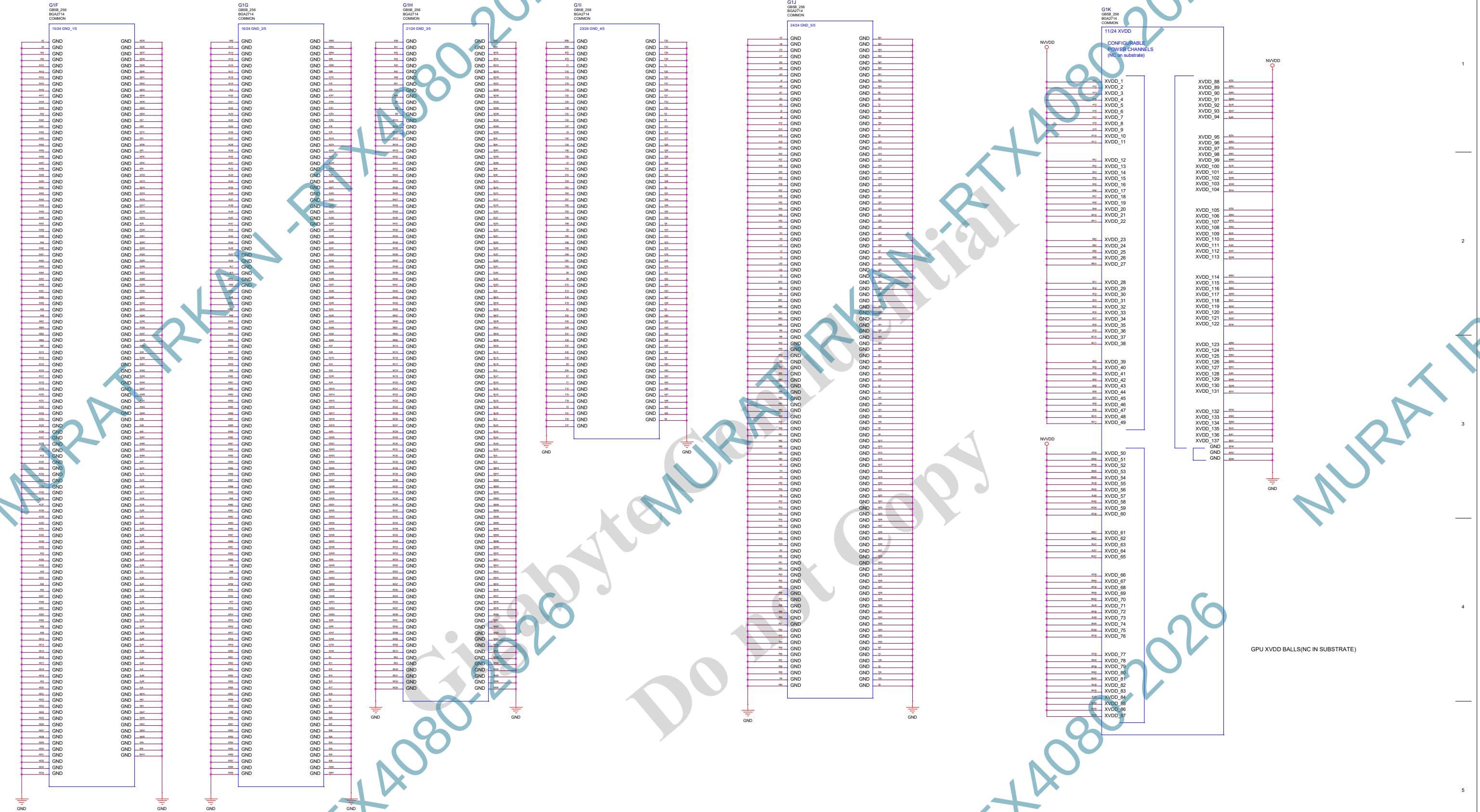
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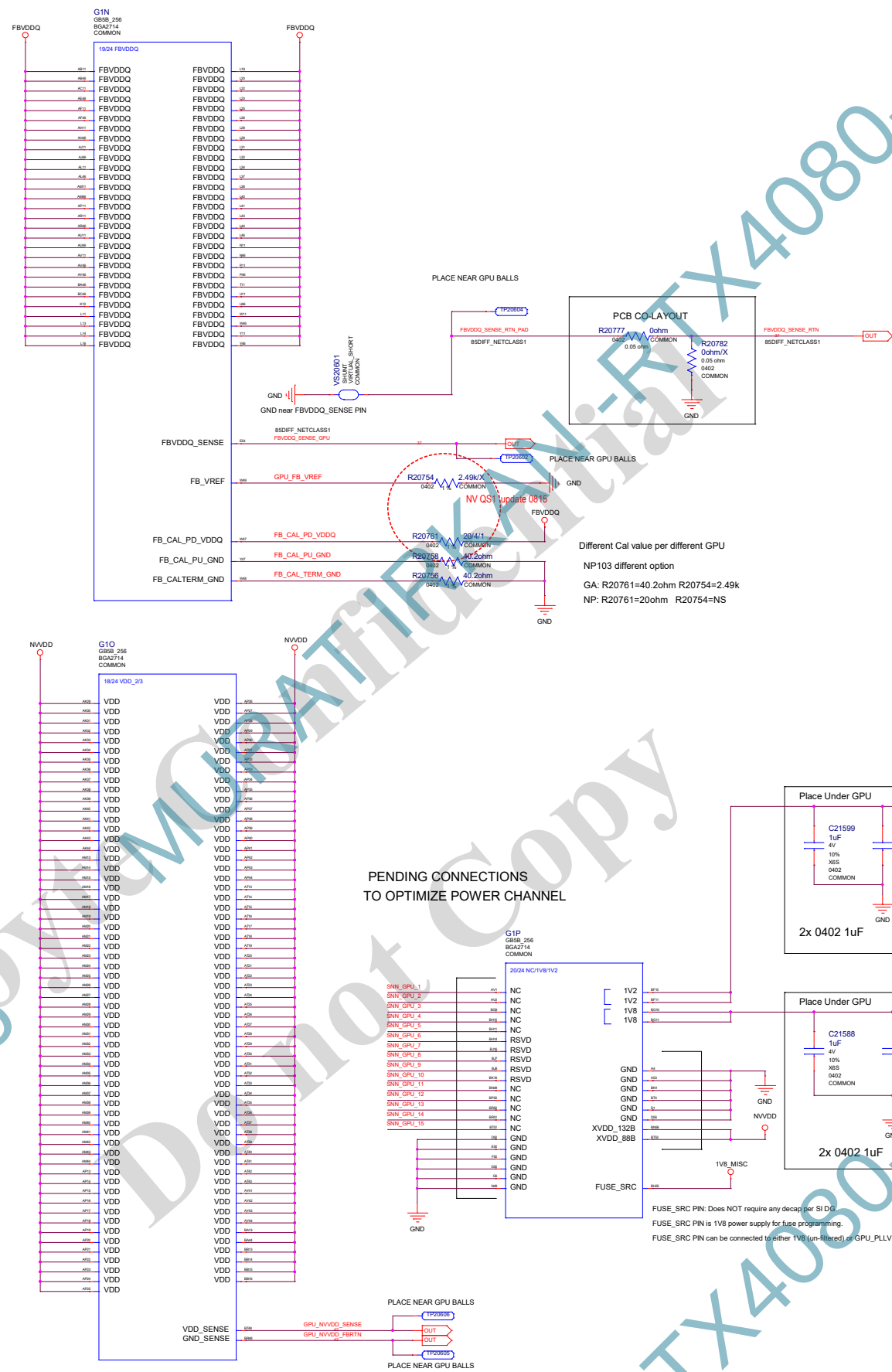
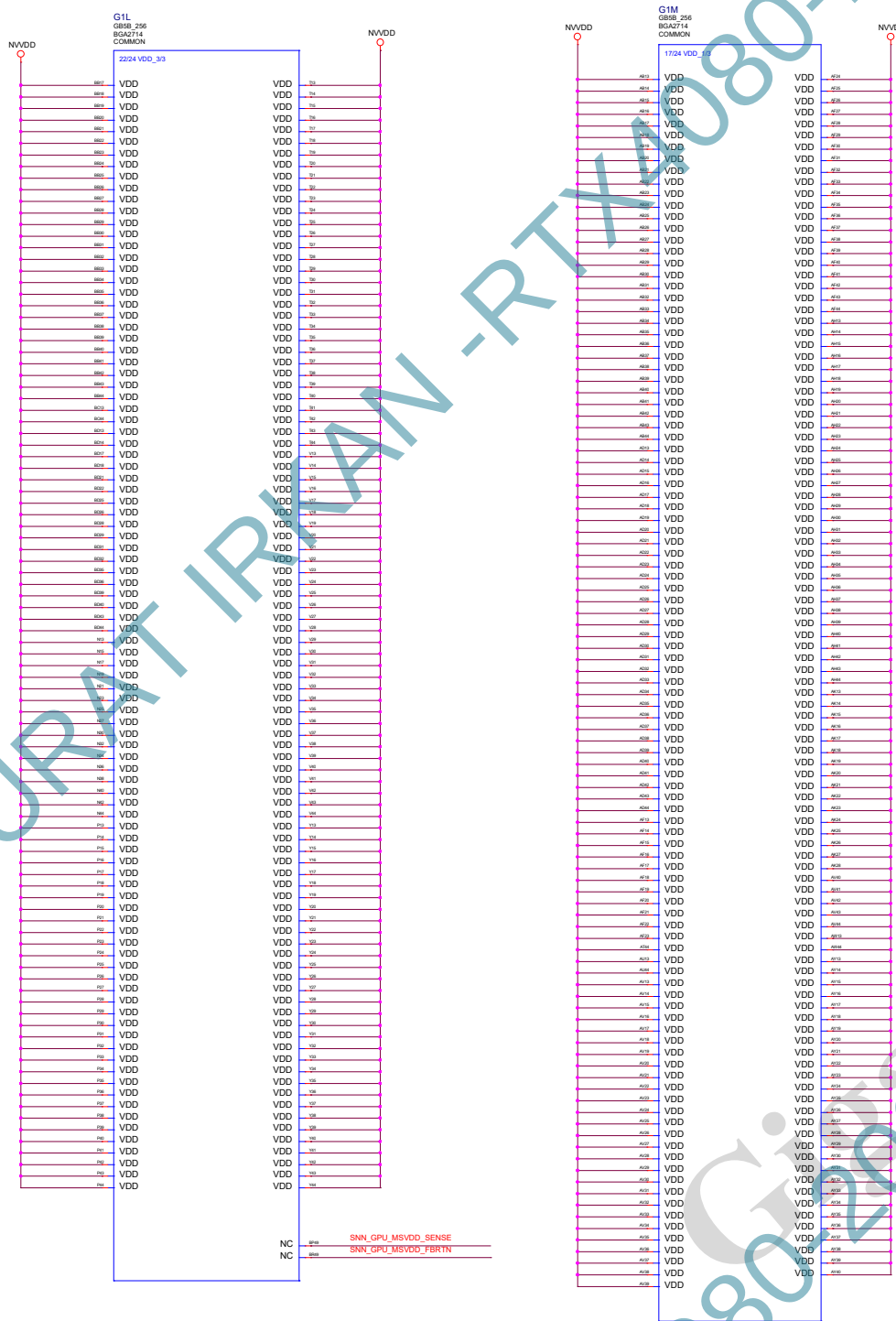
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PER SI REQUIREMENT, RESERVE MORE 0402 1UF AROUND DRAM



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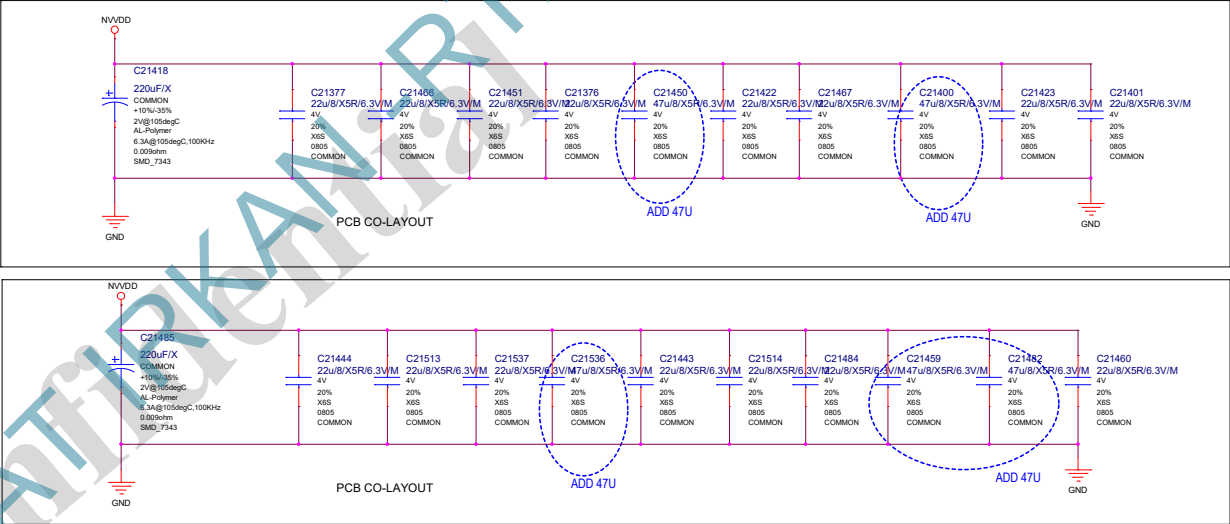
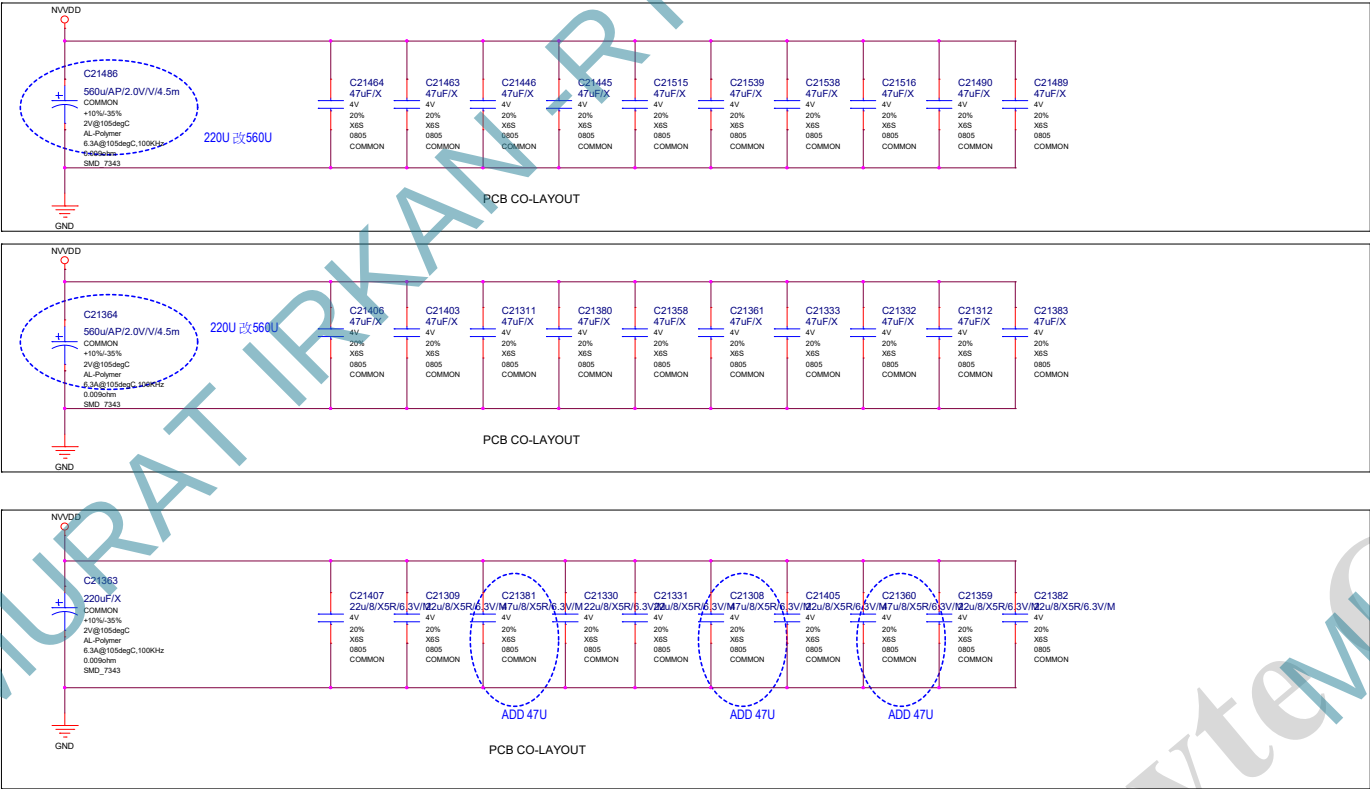


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PAGE DETAIL	GPU: GPU POWER

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NVVDD UNDER GPU



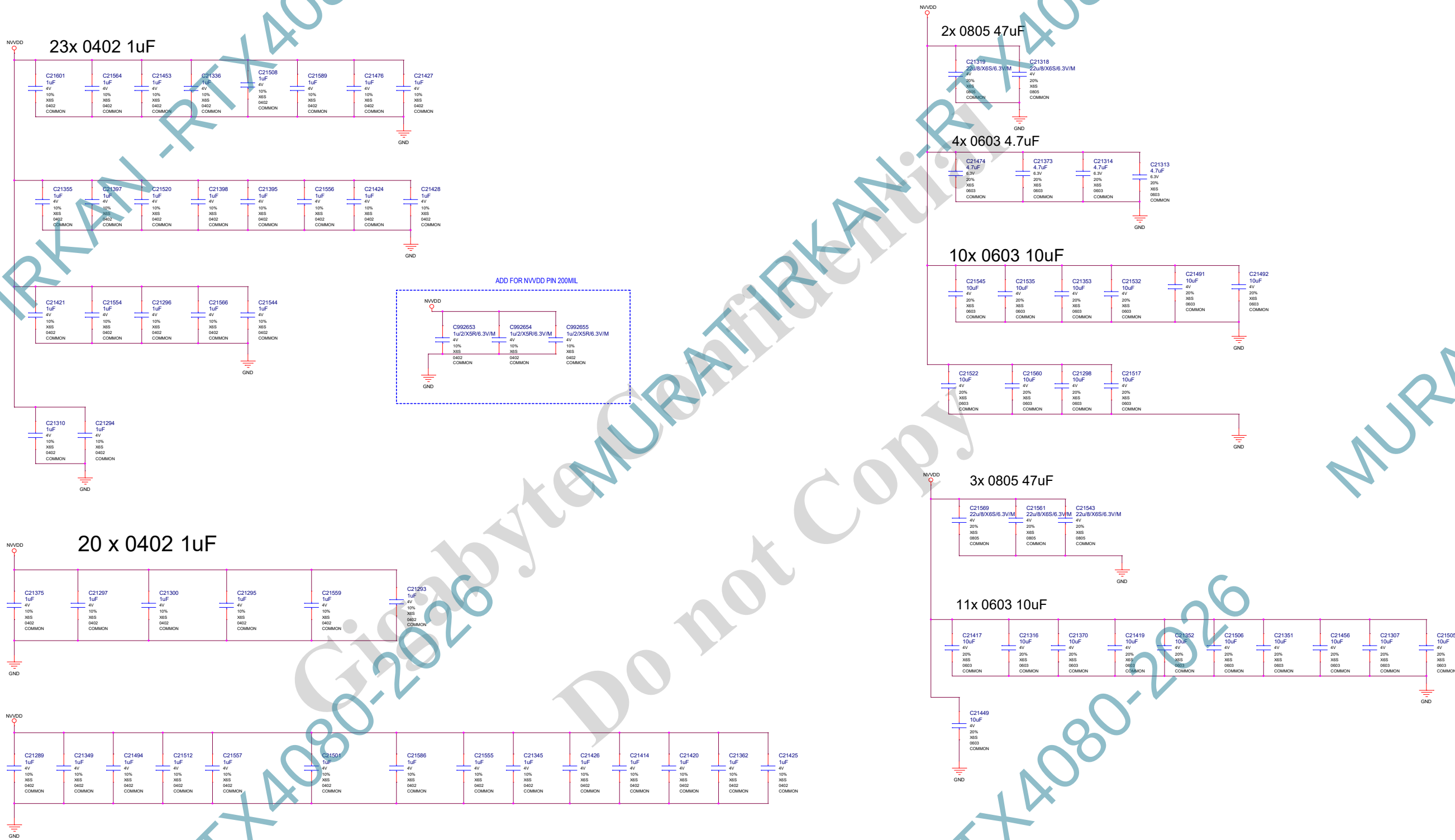
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ASSEMBLY
PAGE DETAIL
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GPU: NVVDD DECOUPLING(1/2)

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NVVDD

UNDER GPU

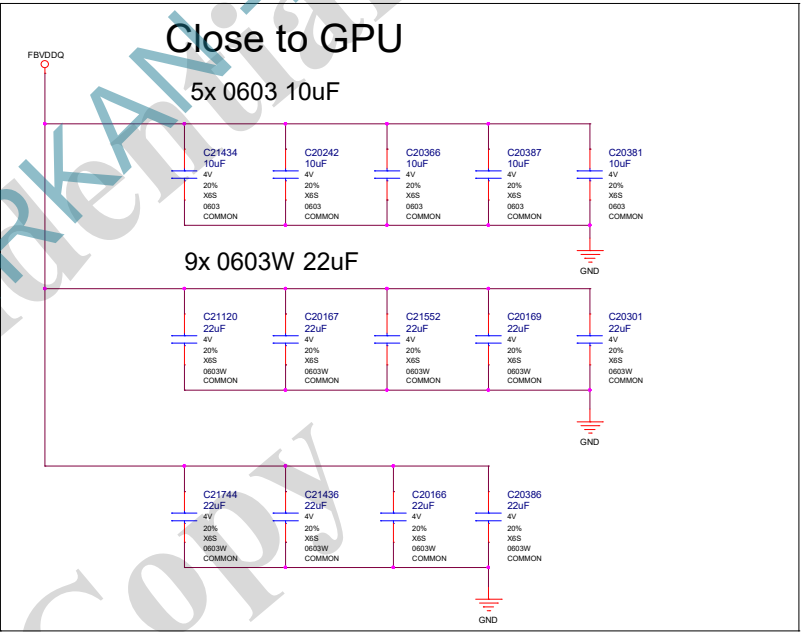
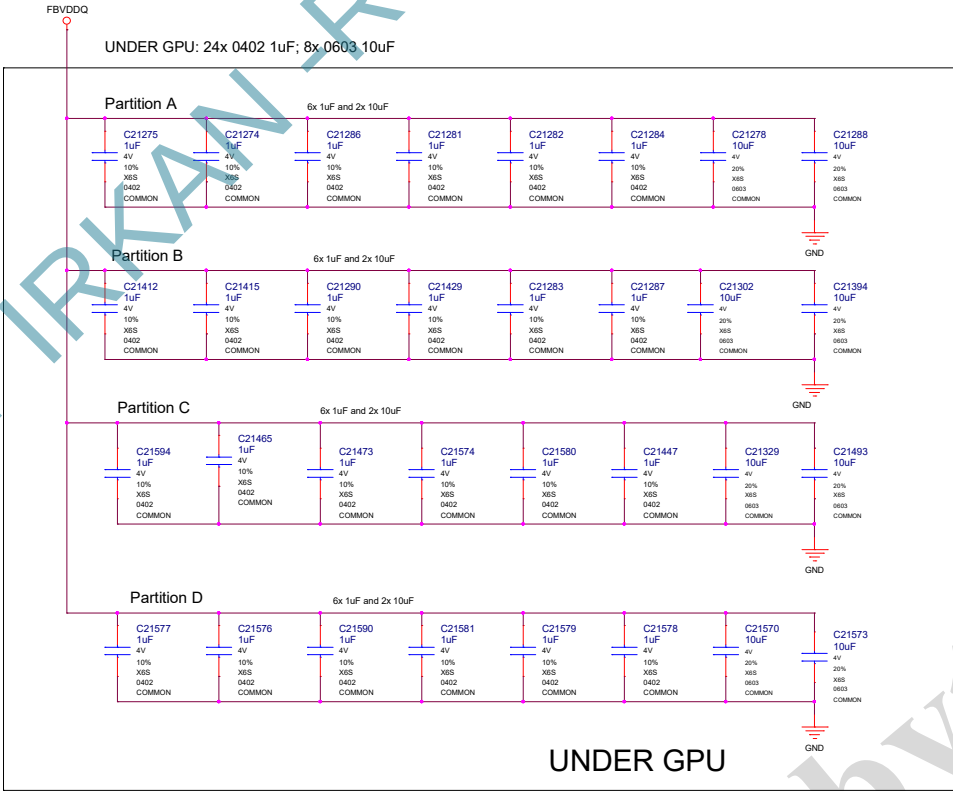


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FBVDDQ



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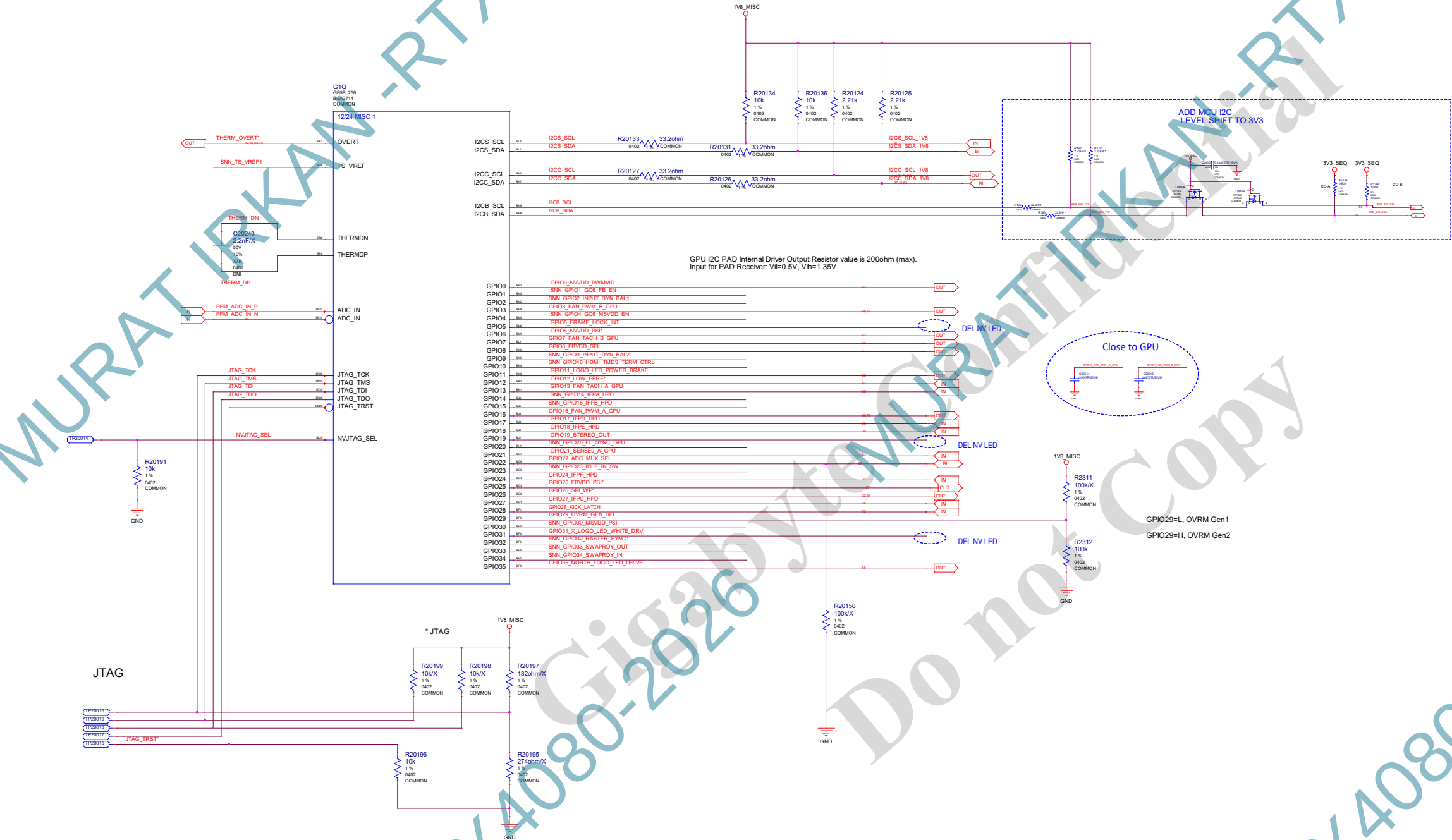
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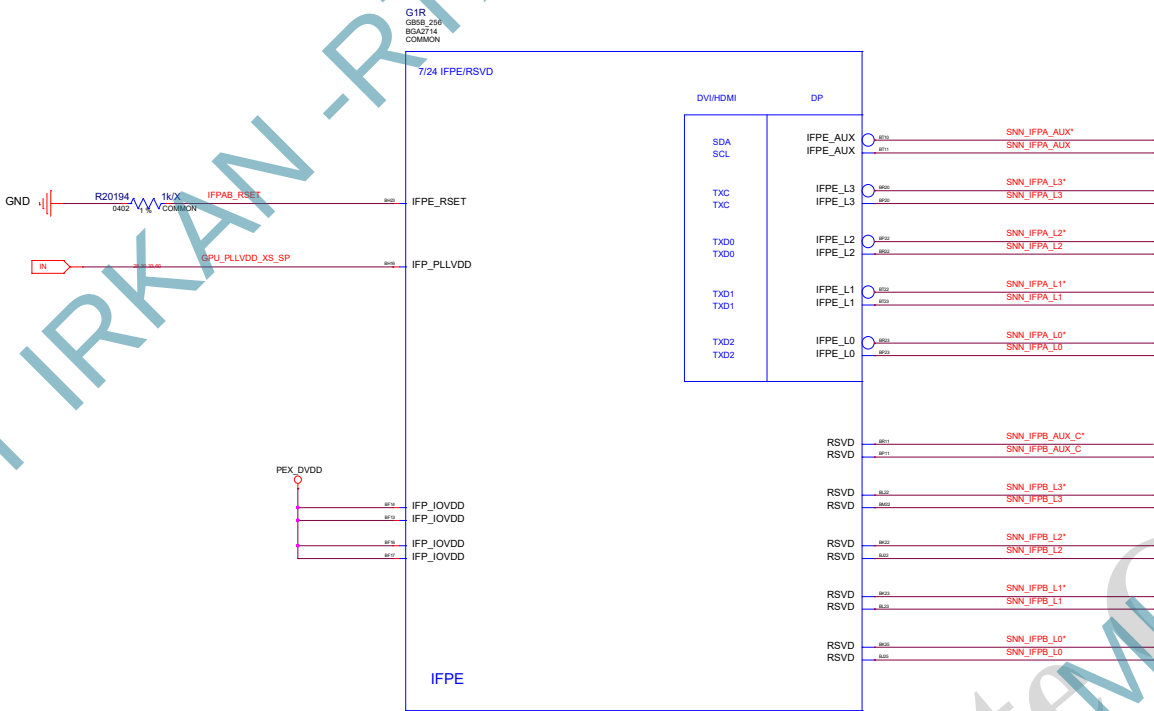
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ASSEMBLY
PAGE DETAIL
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GPU: GPIO, THERMAL, JTAG

GIGABYTE			
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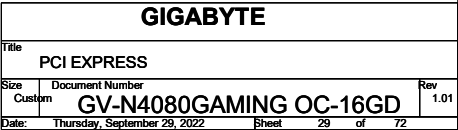


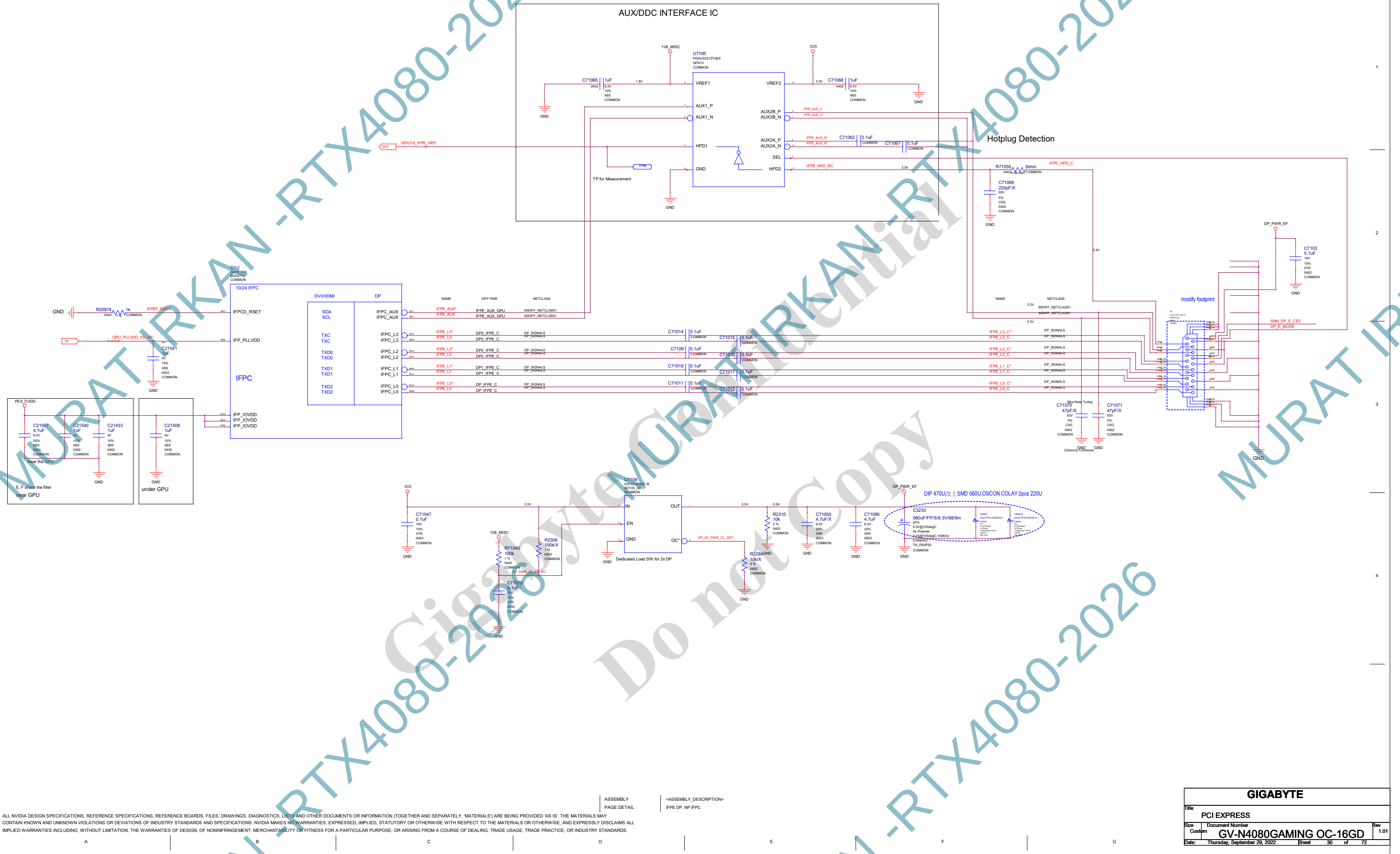
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ASSEMBLY
PAGE DETAIL

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IFPAB#IFPB UNUSED.NP IFPE

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GROUP0			
STRAP2	STRAP1	STRAP0	RAMCFG[4:0]
L	L	H	00001
L	H	L	00010
L	H	H	00011
L	M	H	01010 RAMCFG MICRON 16Gb G6X x16 (161-0479-900)
L	H	M	01011 RAMCFG MICRON 16Gb G6X x16 (161-0480-901)
H	H	H	00111

GROUP1				
ROM_SO	ROM_SI	ROM_SCLK	SMARTFAN[2:0].FS_OVERT	1:ENABLE 0:DISABLE
L	M	L	1001	FAN0/1=33%33%, FS_OVERT ENABLE

GROUP2						
STRAP5	STRAP4	STRAP3	SMB_ALT_ADDR	DEVID_SEL	PCIE_CFG	VGA_DEVICE
M	H	H	1	1	1	1
M	H	L	1	1	1	0
M	L	H	1	1	0	1
M	L	L	1	1	0	0
L	H	M	1	0	1	1
L	M	H	1	0	1	0
L	M	L	1	0	0	1
L	L	M	1	0	0	0
H	H	H	0	1	1	1
H	H	L	0	1	1	0
H	L	H	0	1	0	1
H	L	L	0	1	0	0
L	H	H	0	0	1	1
L	H	L	0	0	1	0
L	L	H	0	0	0	1
L	L	L	0	0	0	0

For NP, Middle Level should be 0.6V
H=High :Tied to 1.8V
M=Middle:Tied to 0.9V
L=Low :Tied to 0V

DEFAULT

<-DEFAULT

- 1:SMB_ALT_ADDR ENABLE
0:SMB_ALT_ADDR DISABLE

1:DEVID_SEL REBRAND
0:DEVID_SEL ORIGINAL

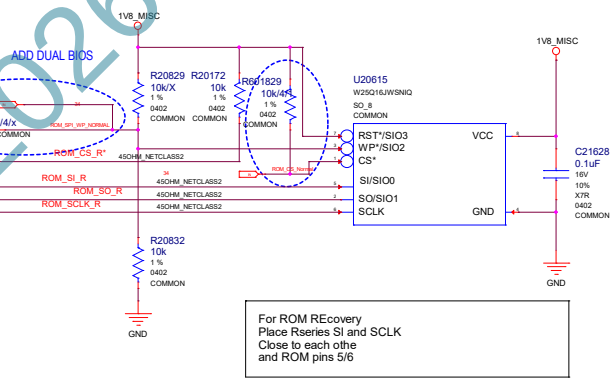
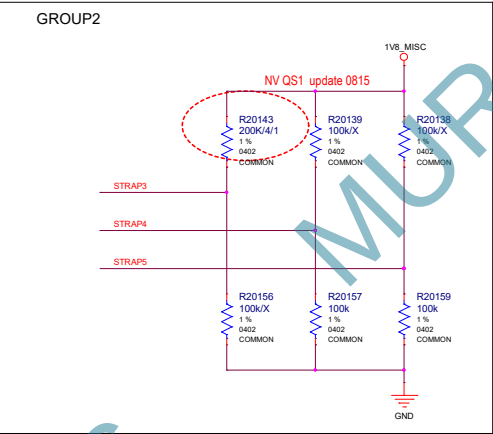
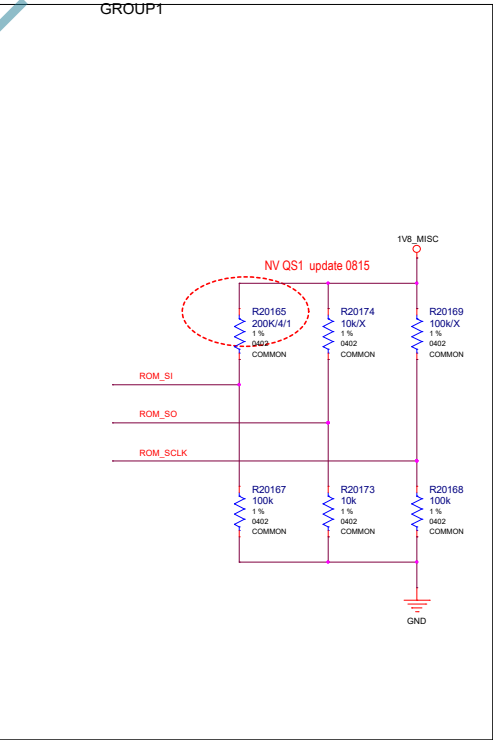
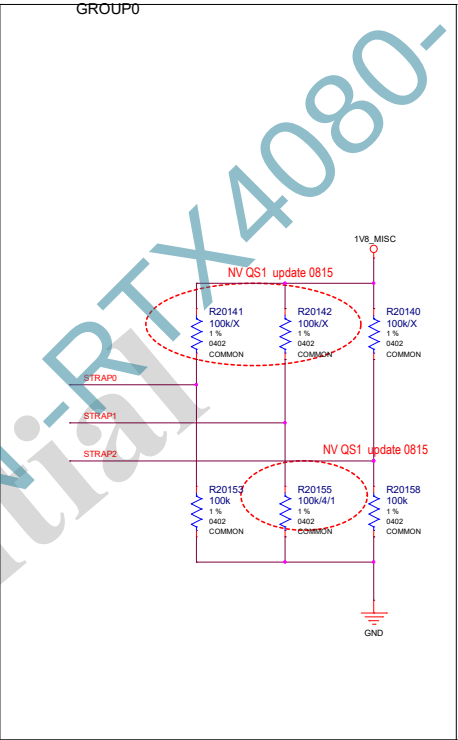
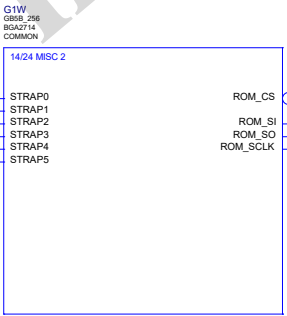
1:PCIE_CFG LOW POWER
0:PCIE_CFG HIGH POWER

1:VGA_DEVICE ENABLE
0:VGA_DEVICE DISABLE

Default

SKU 200

STRAP0
STRAP1
STRAP2
STRAP3
STRAP4
STRAP5

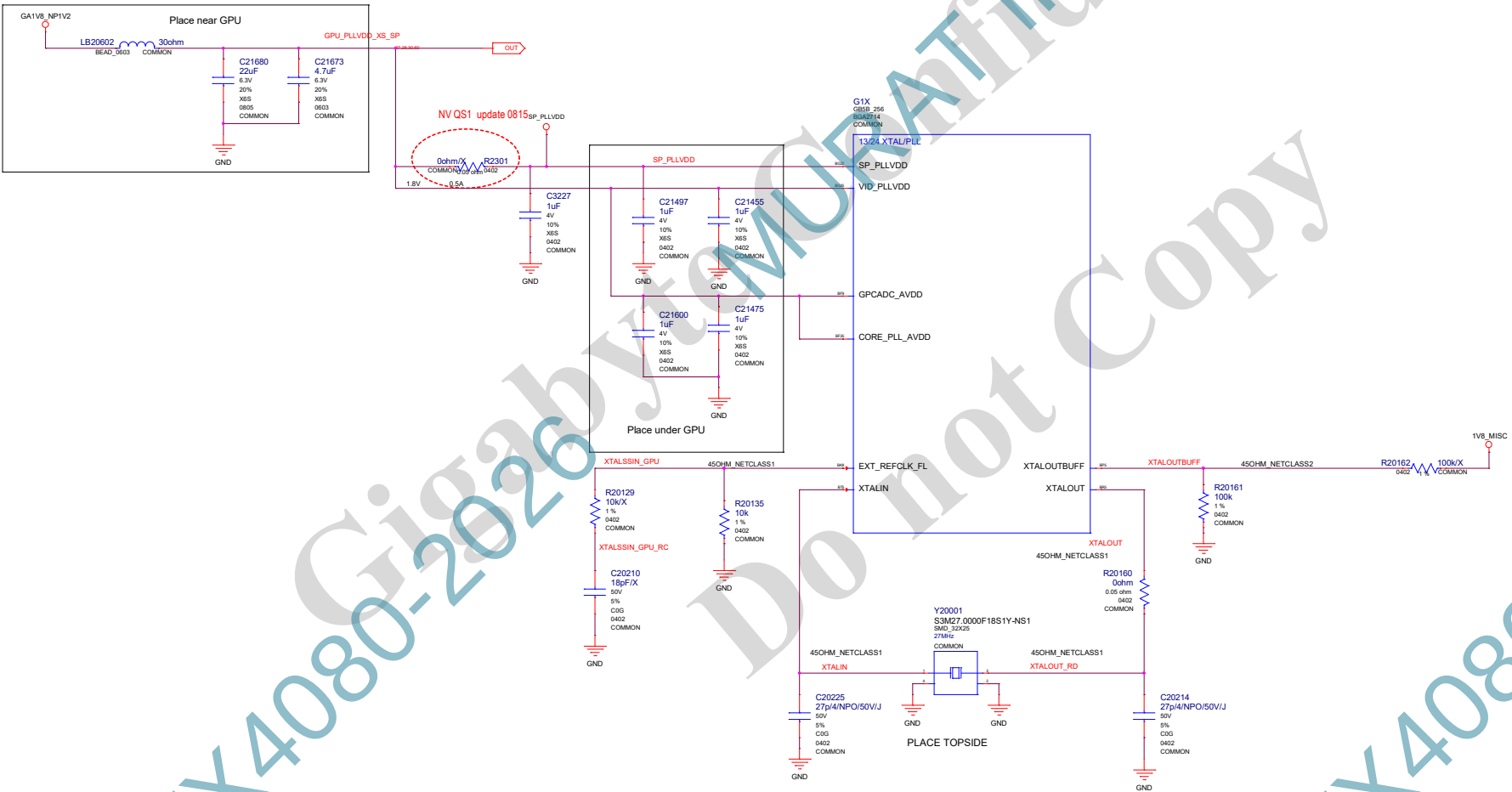


For ROM REcovery
Place Rseries SI and SCLK
Close to each othe
and ROM pins 5/6

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ASSEMBLY
PAGE DETAIL
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MISC. ROM, STRAPS

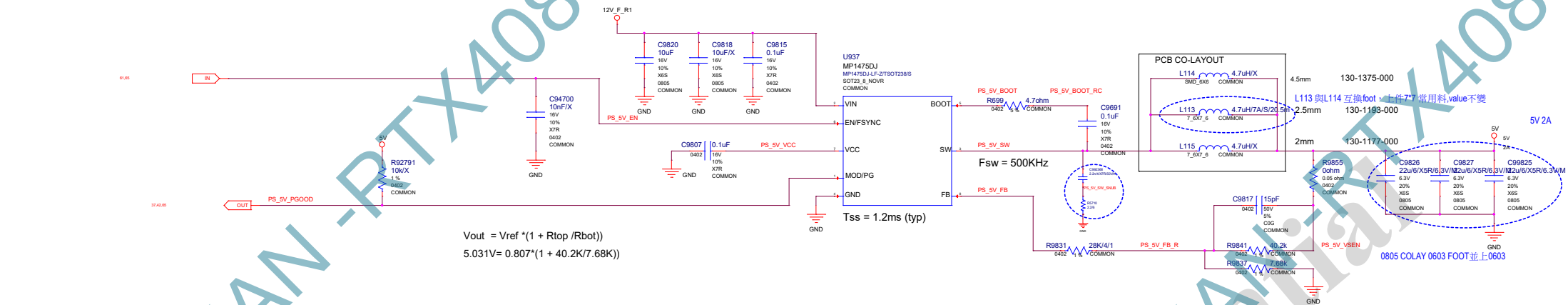
GIGABYTE			
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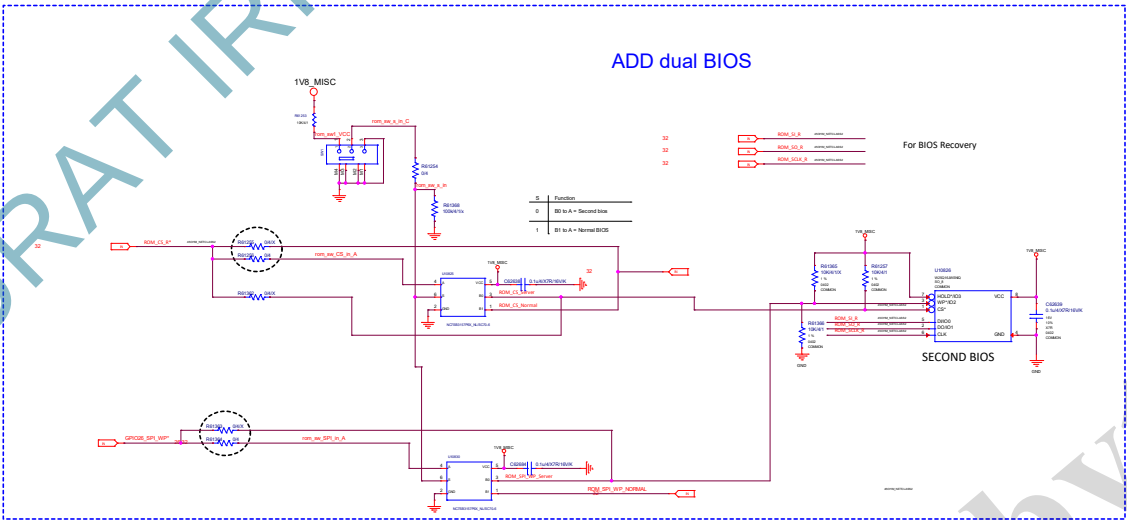
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ASSEMBLY
PAGE DETAIL
<ASSEMBLY_DESCRIPTION>
MISC. XTAL. PLL

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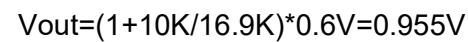
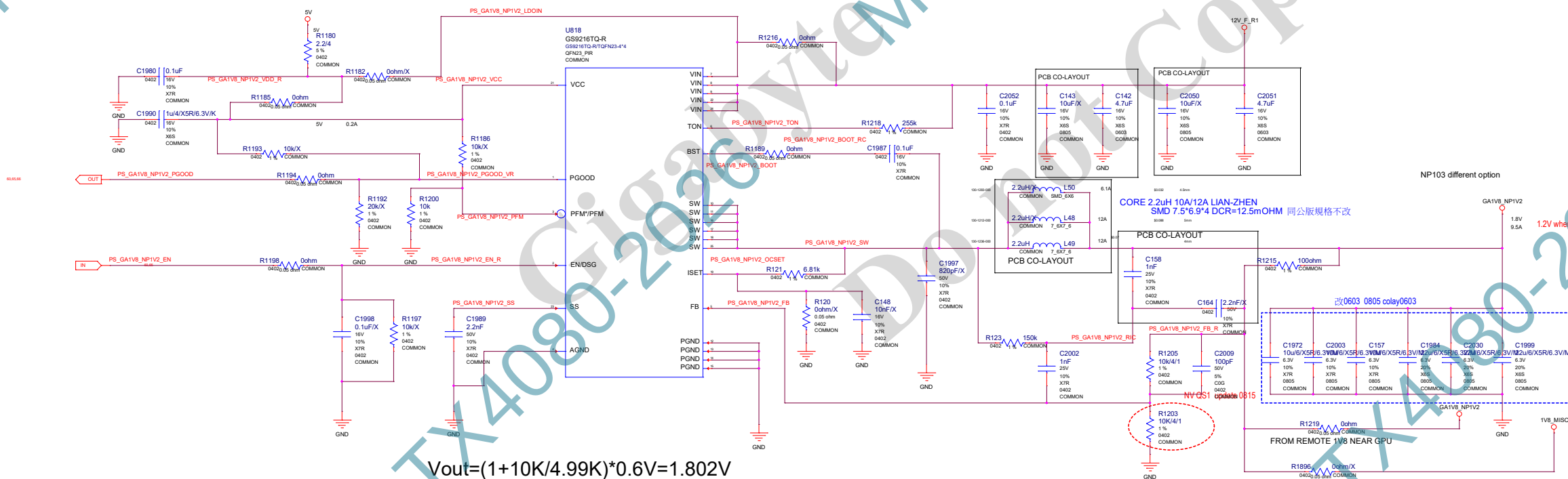
$$V_{out} = V_{ref} * (1 + R_{top} / R_{bot})$$
$$5.031V = 0.807 * (1 + 40.2K / 7.68K)$$



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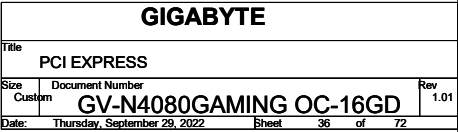
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PS: 5V

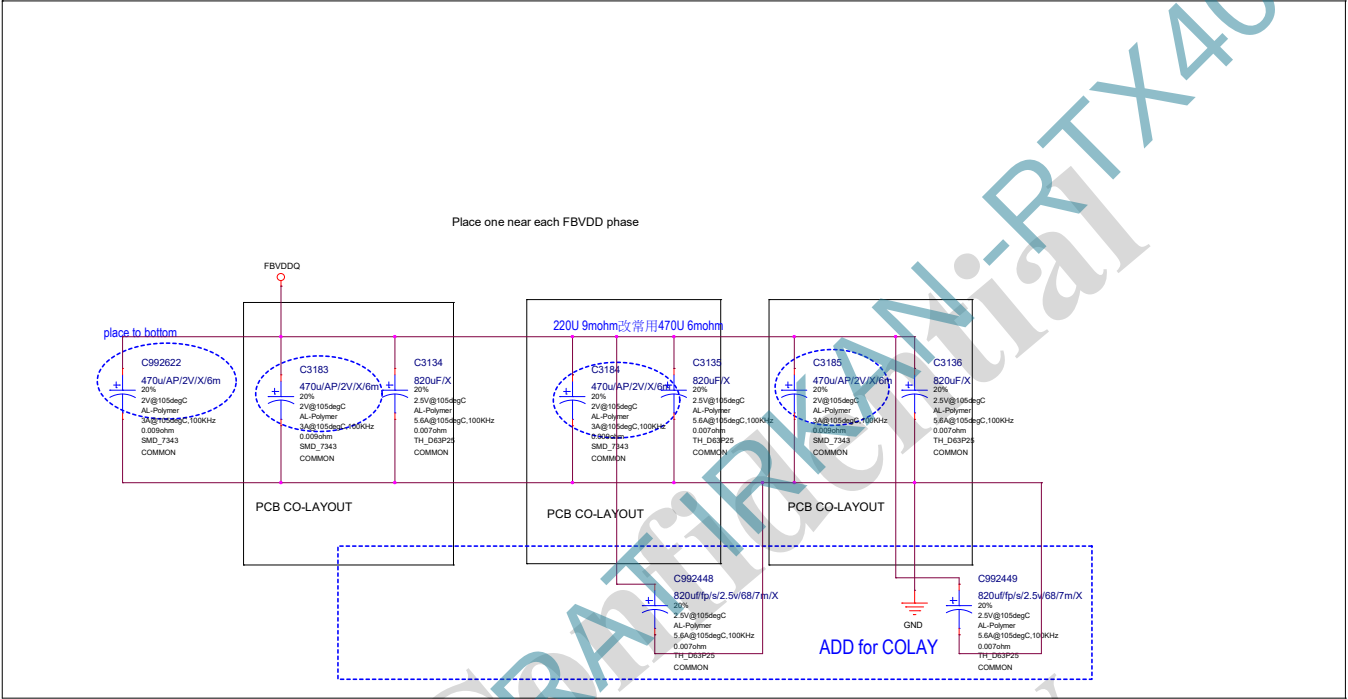
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Size			
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$$V_{out} = (1 + 10K/4.99K) * 0.6V = 1.802V$$


ASSEMBLY	<ASSEMBLY_DESCRIPTION>
PAGE DETAIL	PS: PEX_DVDD, GA1V8_NP1V2

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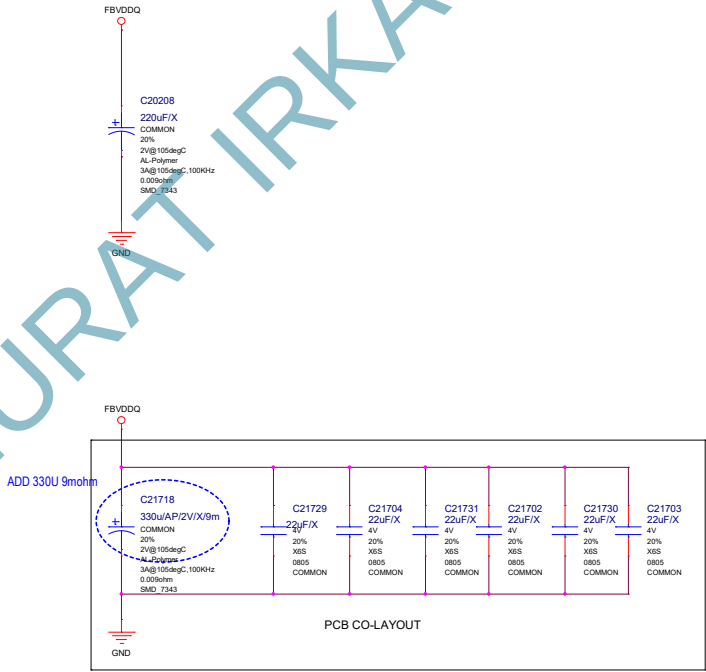


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ASSEMBLY
PAGE DETAIL

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PS: FBVDD OUTPUT CAP NEAR VR

GIGABYTE				
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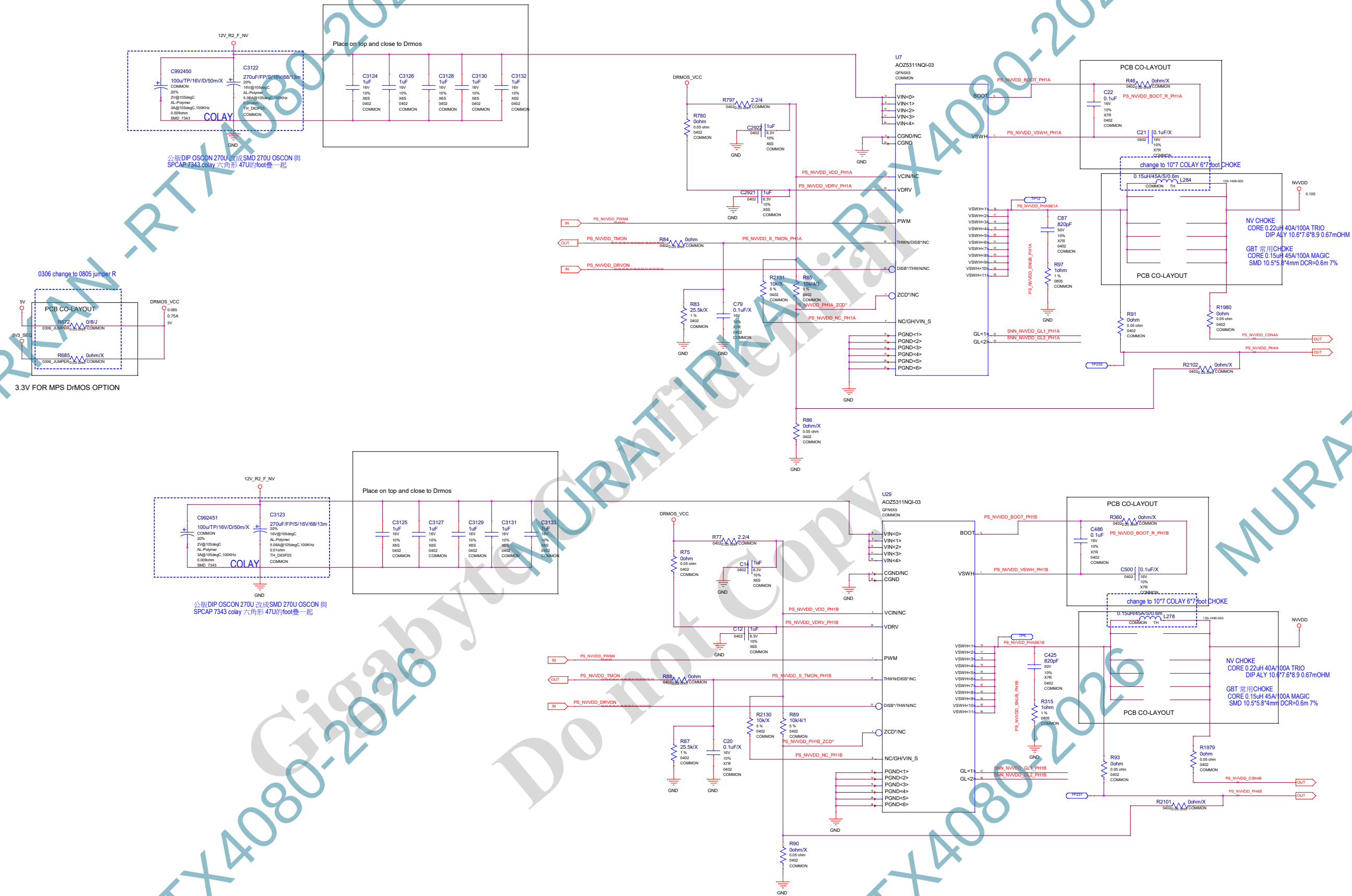


PLACE NEAR MEMORY

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ASSEMBLY PAGE DETAIL
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PS: FBVDD OUTPUT CAP NEAR MEMORY

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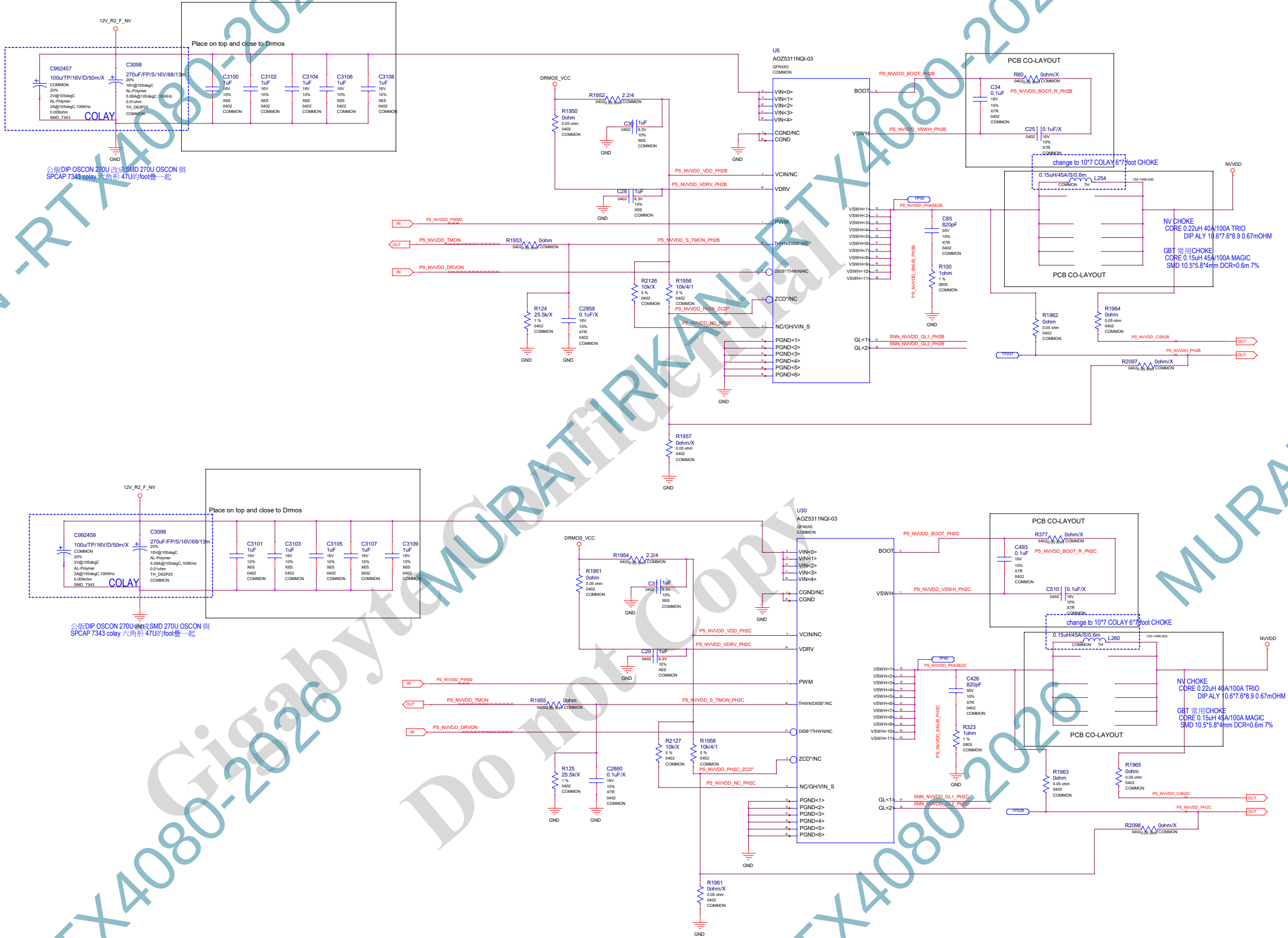


Stuff for AORUS

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ASSEMBLY
PAGE DETAIL
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PS: NVVDD PH1C, PH2A

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ASSEMBLY
PAGE DETAIL
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PS: NVVDD PH2B, PH2C

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Page 48: PS: NVVDD PH3C,PH4A

Remove Phase 4A for 18PHASE 20220914

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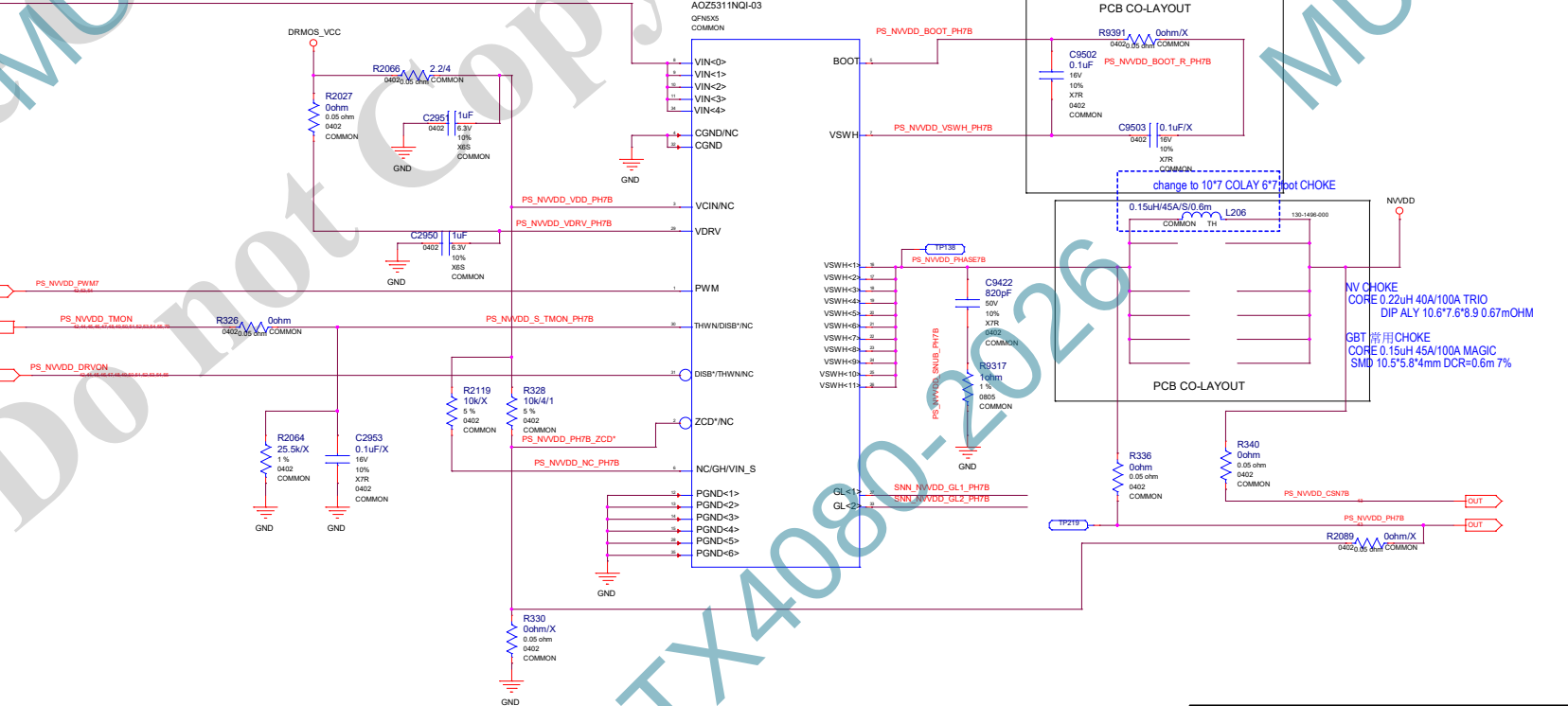
Page 48: PS: NVVDD PH3C,PH4A

Remove Phase 4A for 18PHASE 20220914

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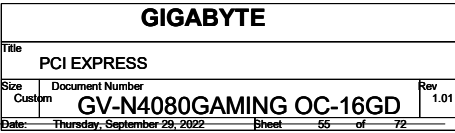


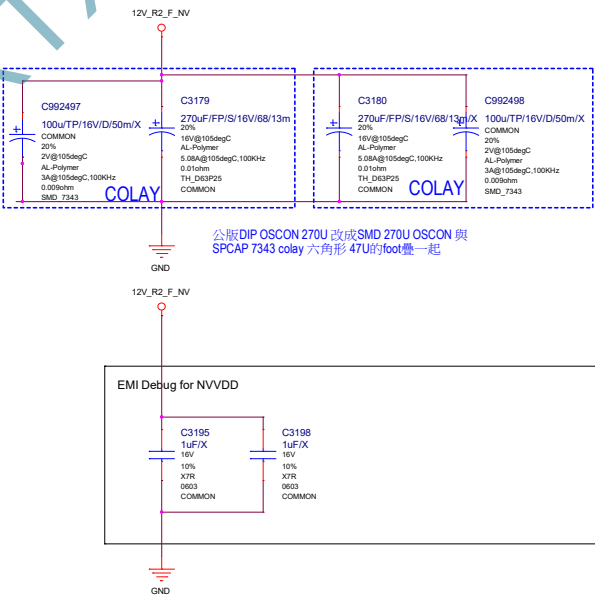
公版DIP OSCON 270U 改或SMD 270U OSCON 與
SPCAP 7343 colay 六角形 47U的foot疊一起



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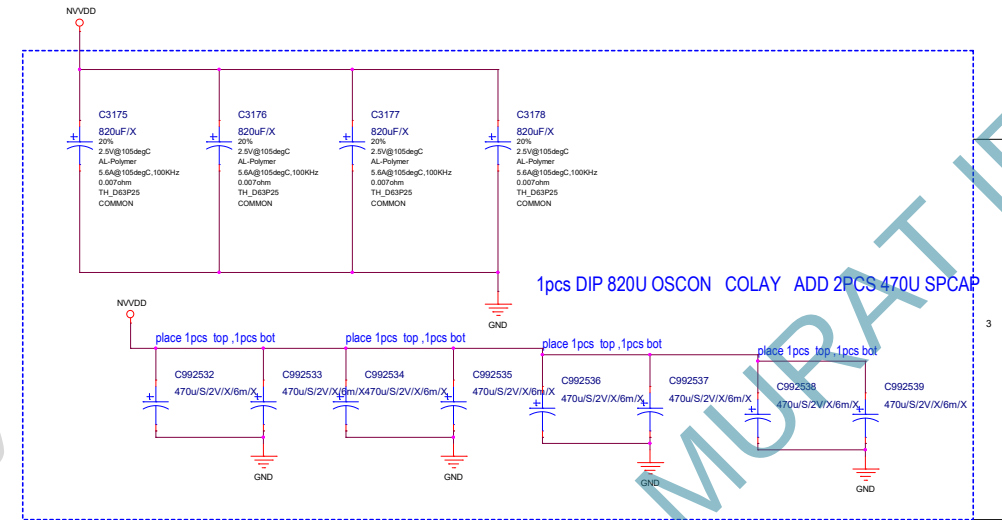
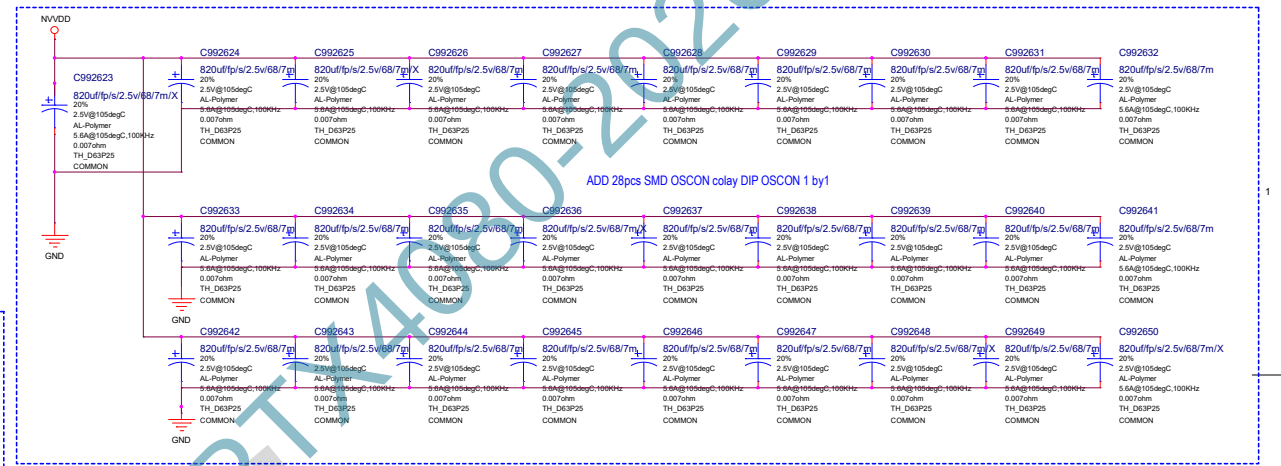


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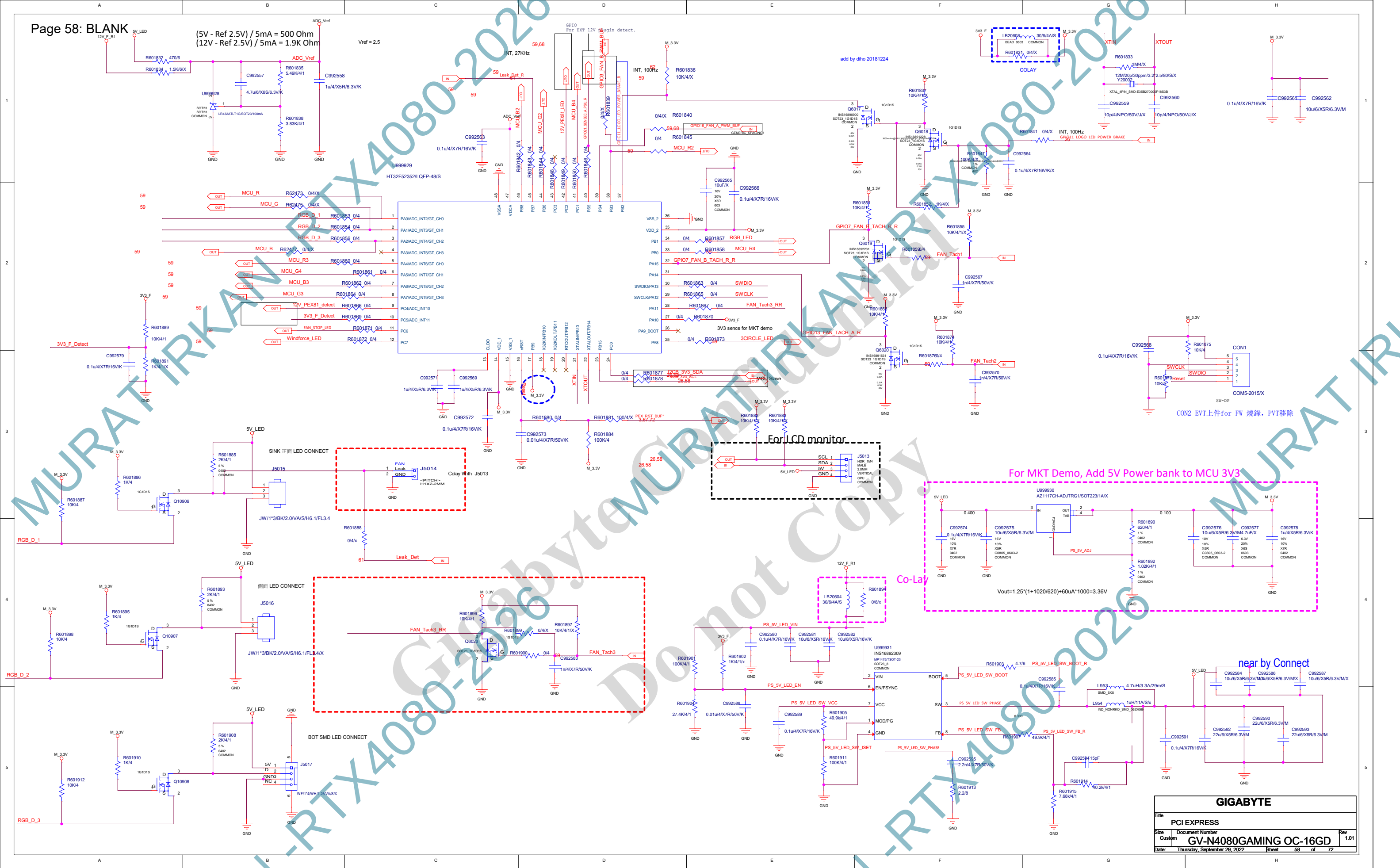
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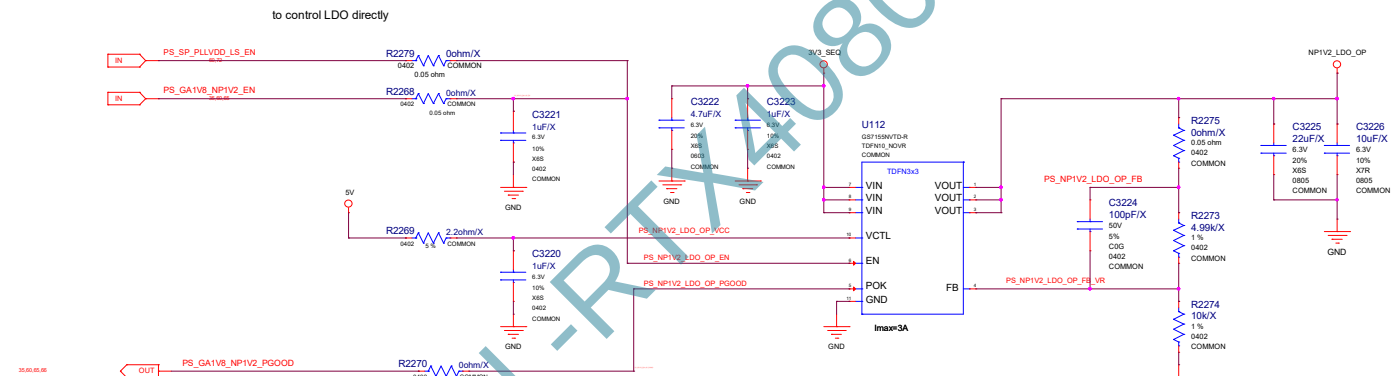
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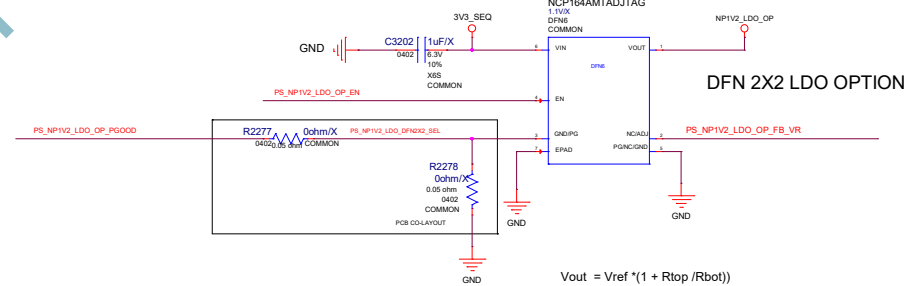
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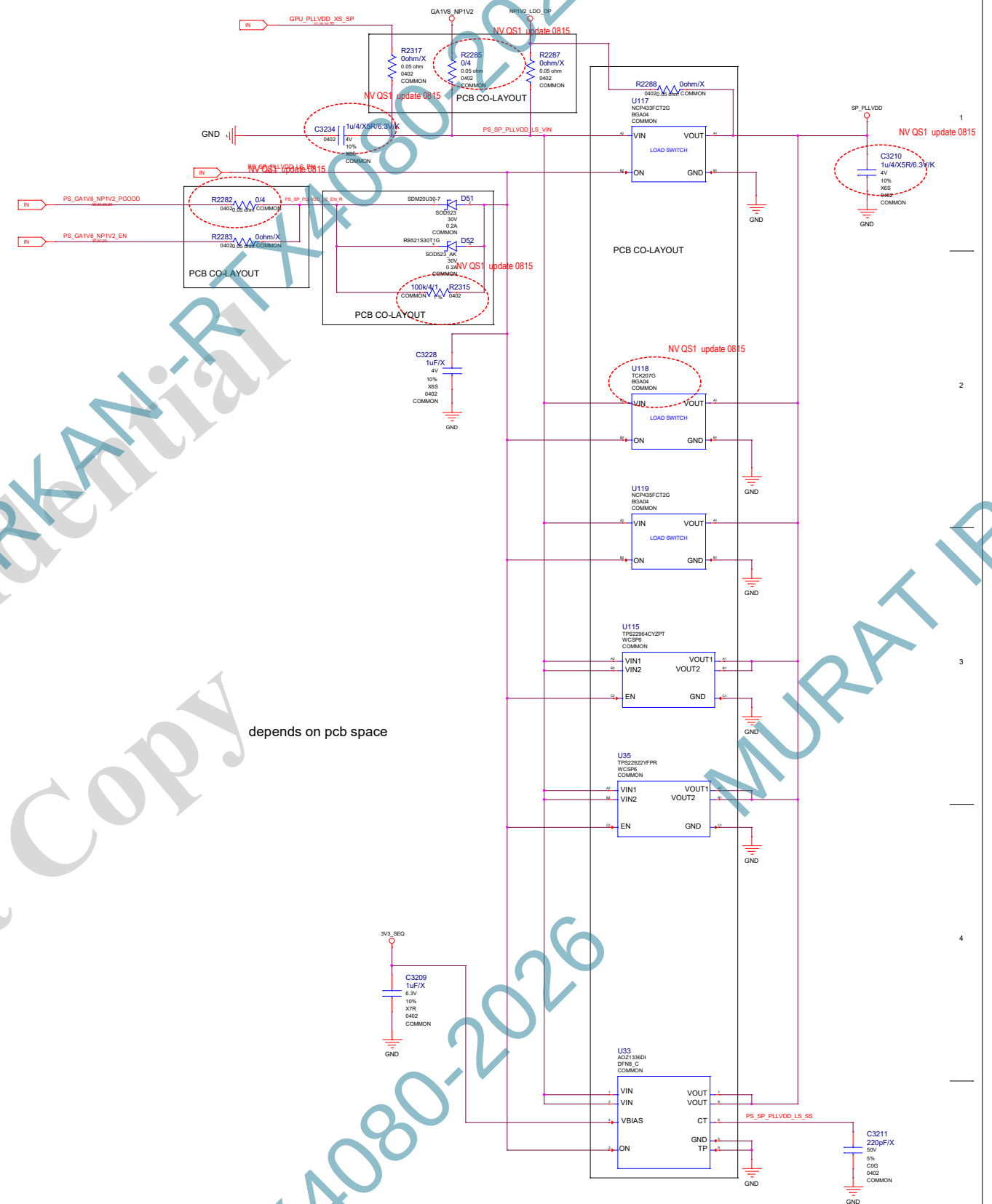
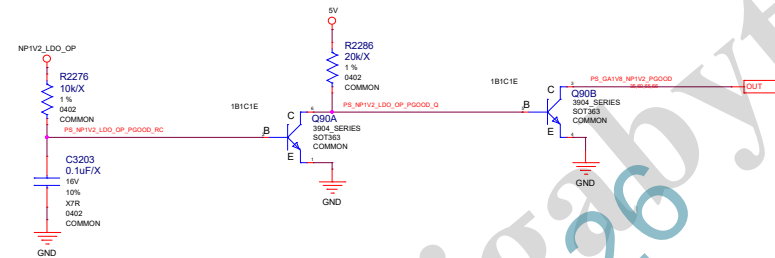
$$1.2V = 0.8V \cdot (1 + 4.99K/10K)$$

315-1283-000 ,NCP164 ADJ Voltage, PGOOD
315-1395-000 ,NCP164 Fix Voltage, PGOOD
315-1067-000 ,NCP705 Fix Voltage, NO PGOOD, PIN3=GND



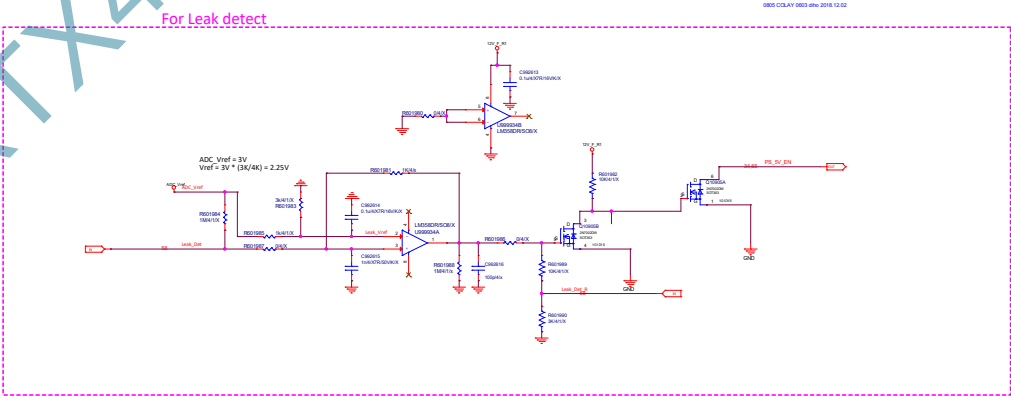
$$V_{out} = V_{ref} * (1 + R_{top} / R_{bot})$$

$$1.2V = 1.1 * (1 + 1K / 11K)$$



depends on pcb space

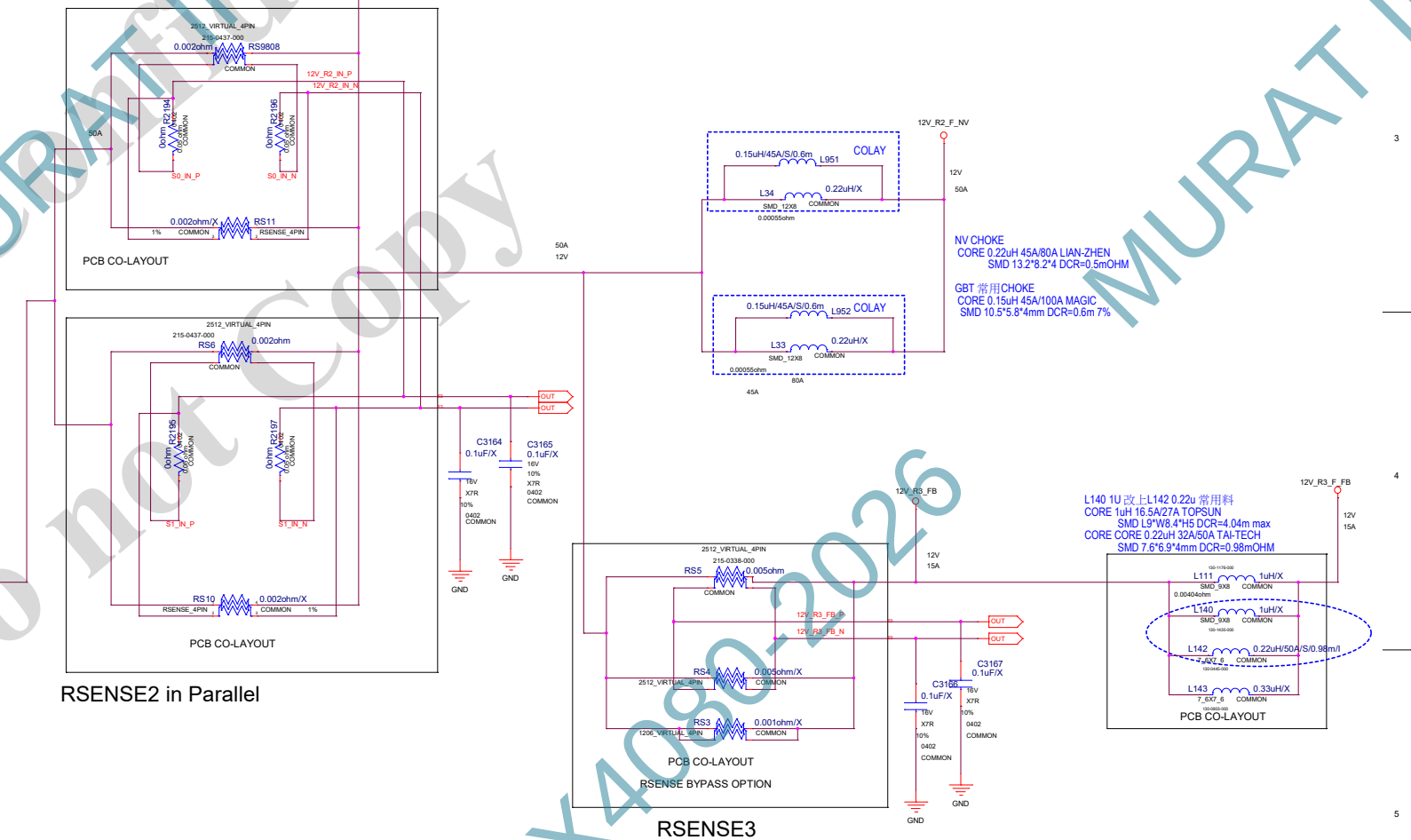
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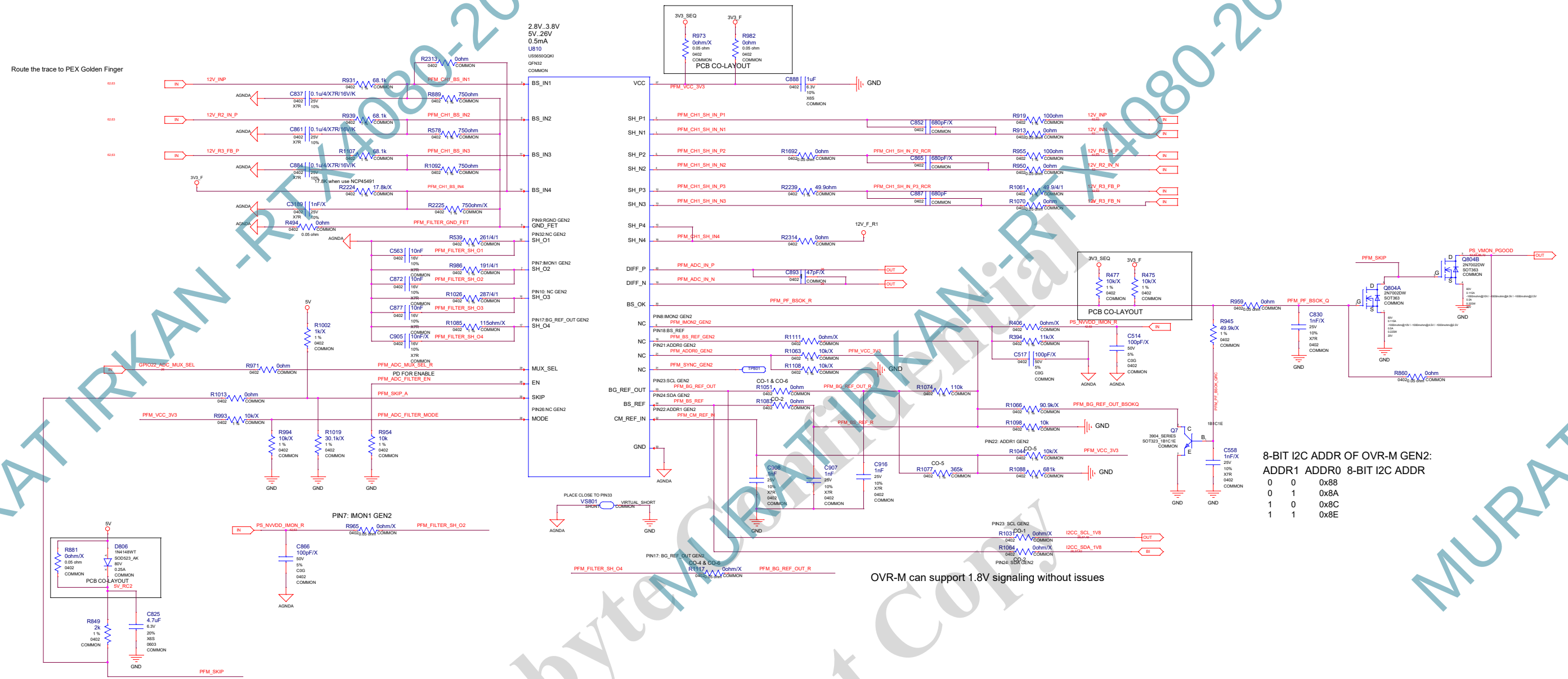
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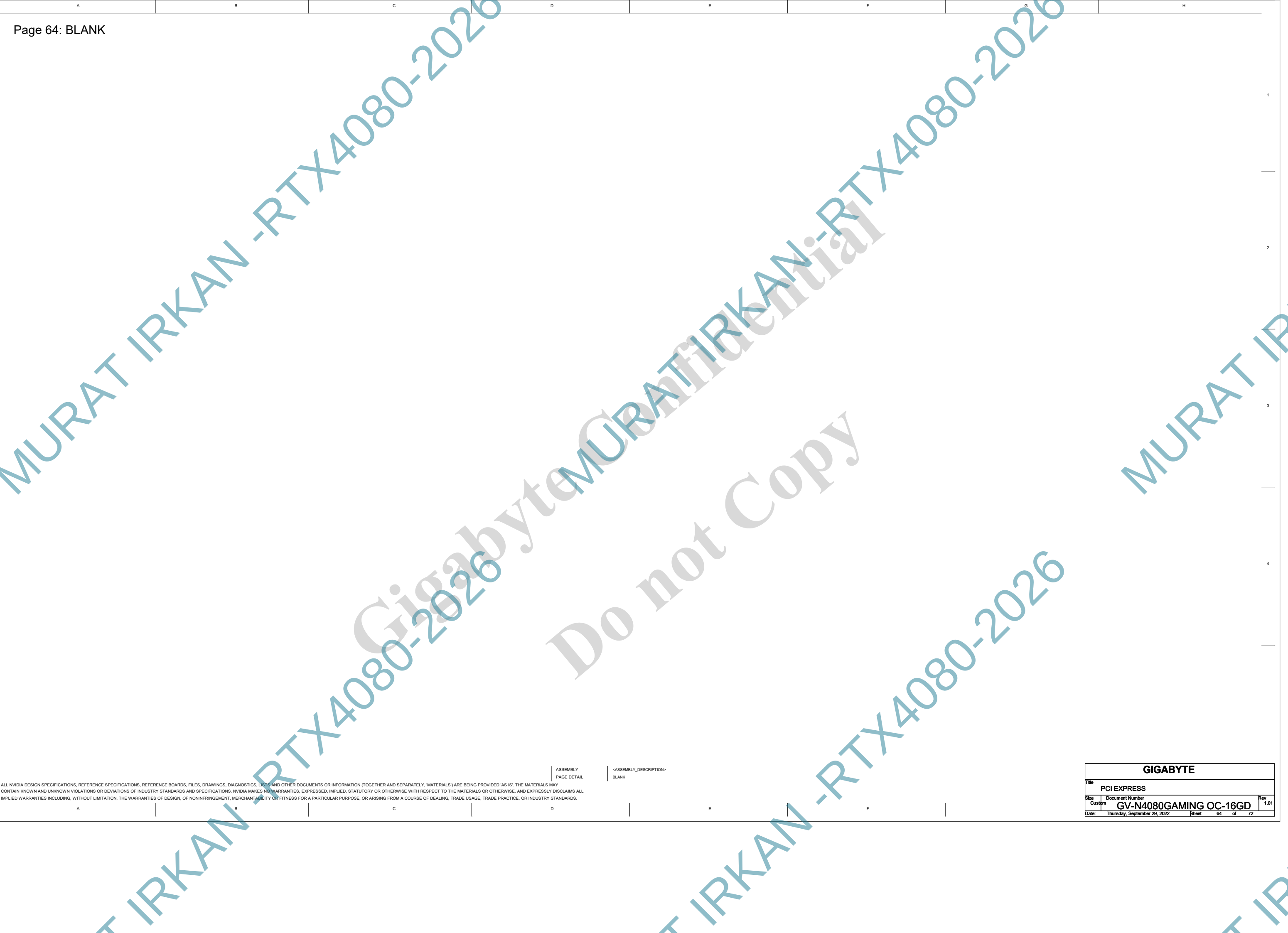
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OVR-M GEN2 SUPPORT

OVR-M GEN2:
PIN 7, 8, 9, 10, 17, 18, 21, 22, 23, 24, 26, 32
are different from GEN1 OVRM,

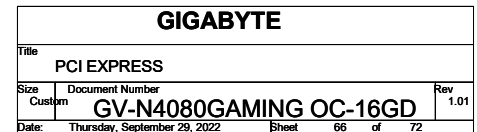
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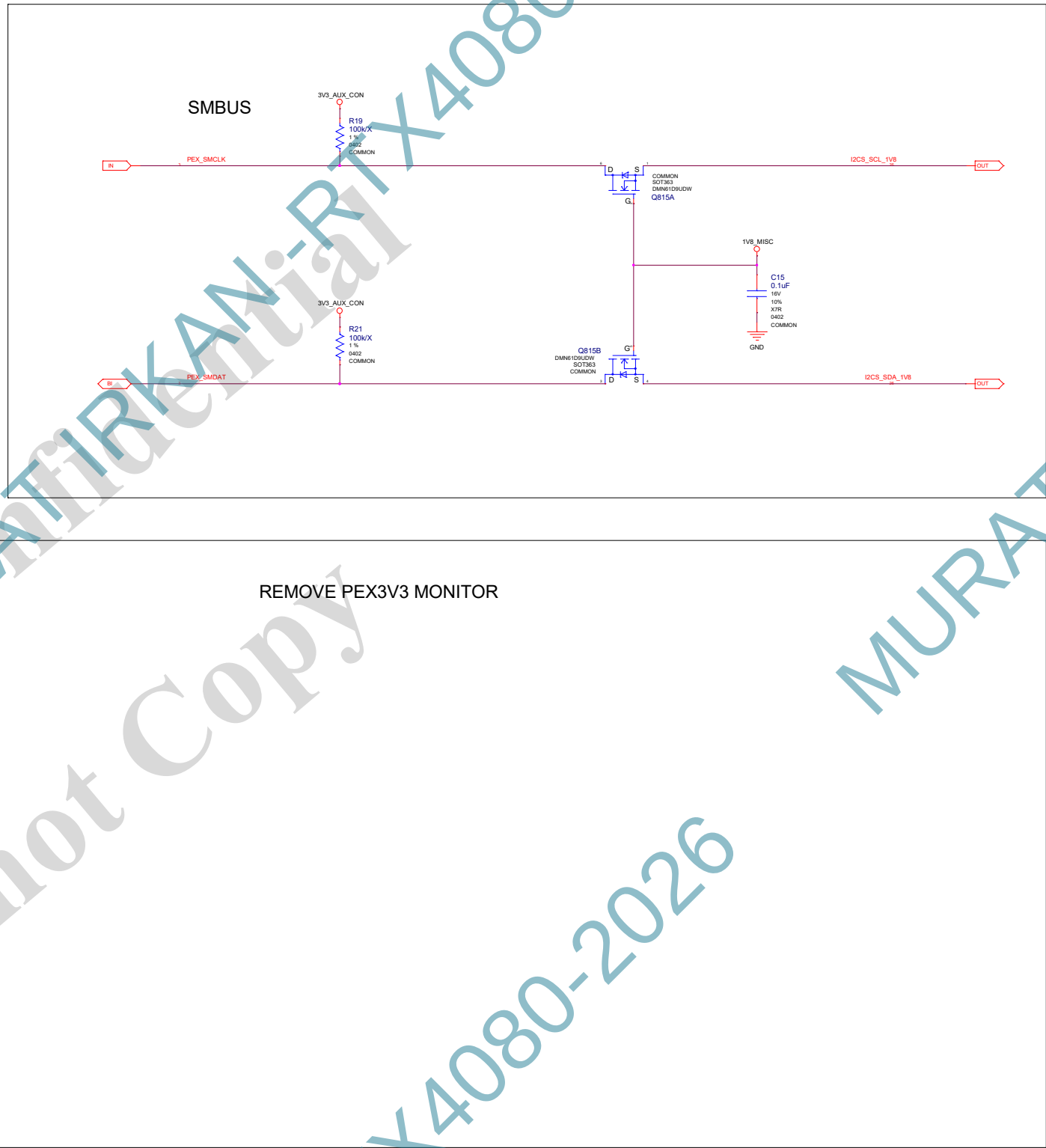
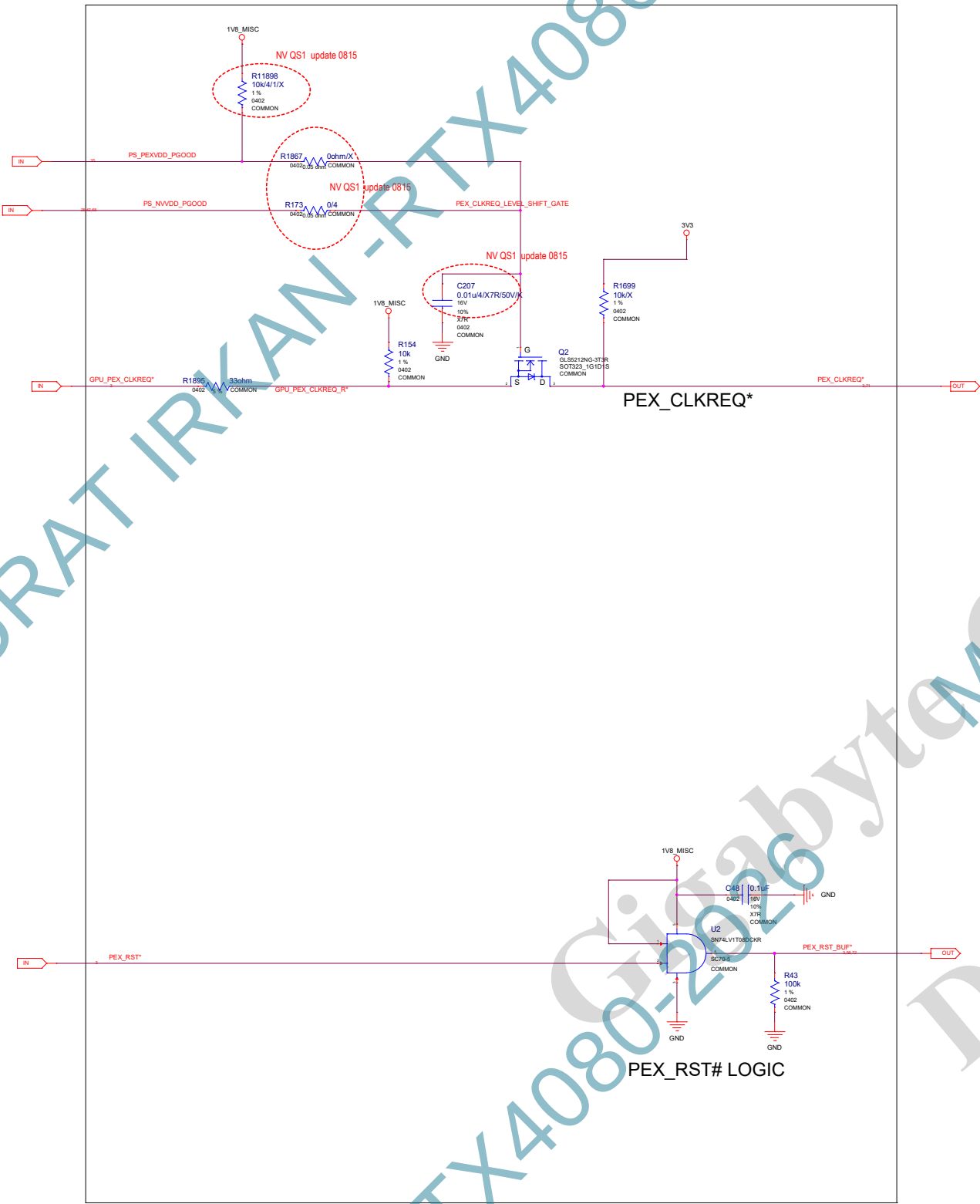


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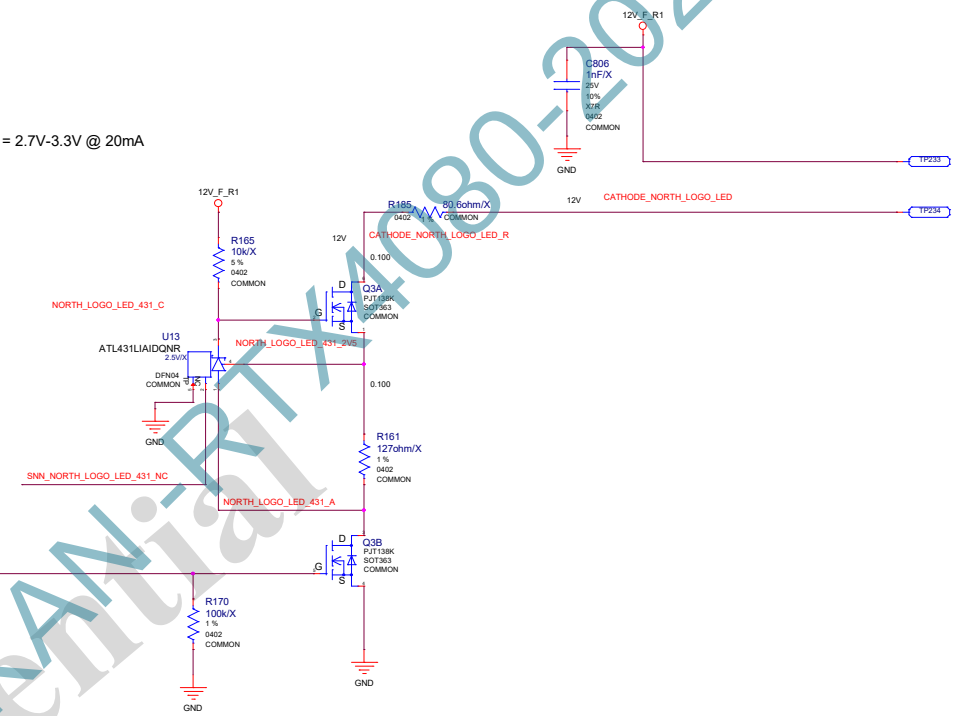
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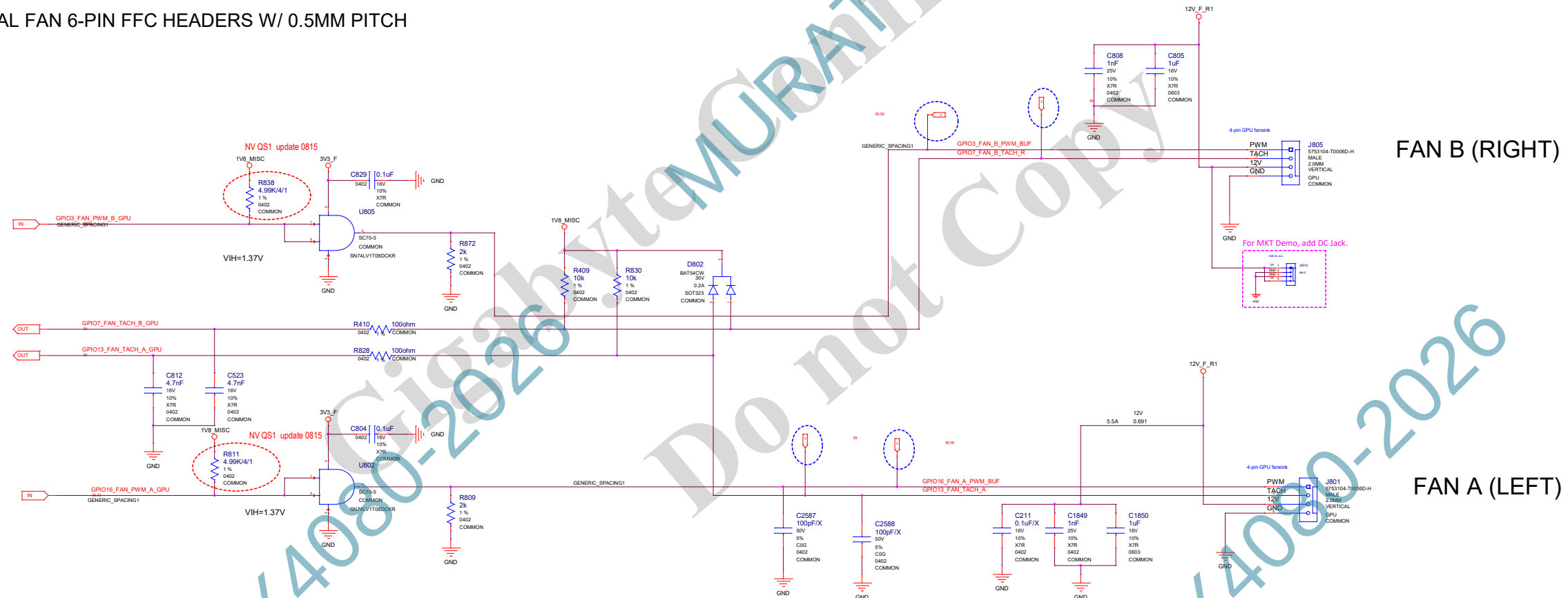
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NORTH LOGO LED (WHITE)

(LED should not blink during boot)



FAN B (RIGHT)



FAN A (LEFT)

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'X' LOGO LED WHITE (3PCS)

(LED should not blink during boot)

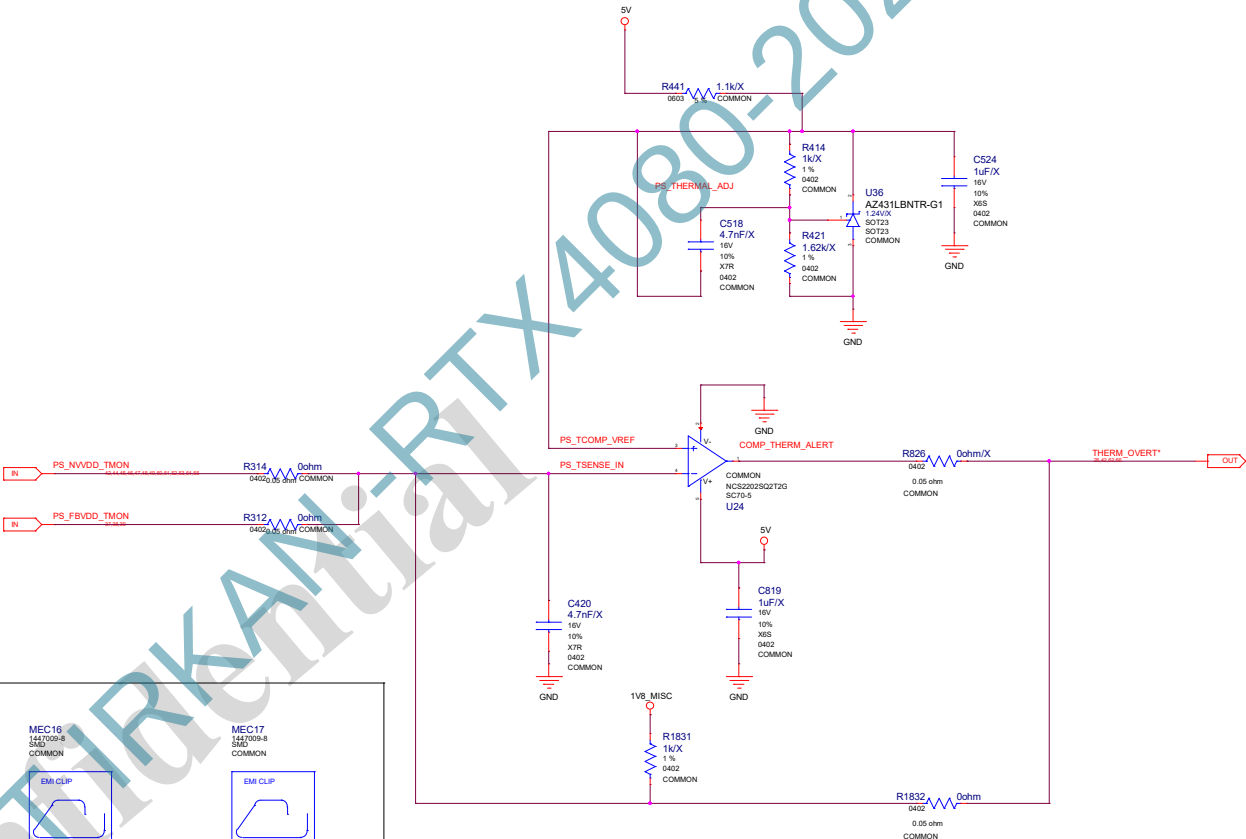
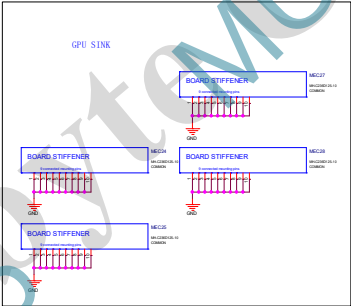
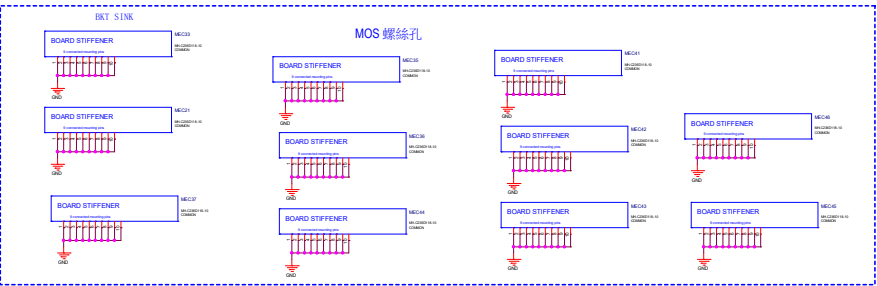
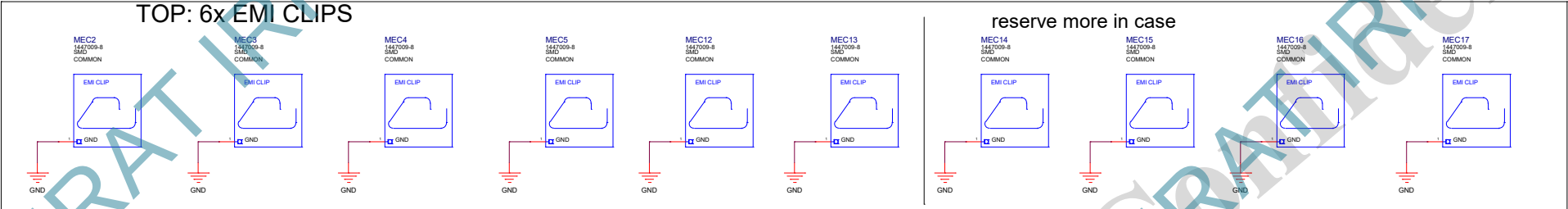
'X' LOGO LED RED (3PCS)

/X LED線路

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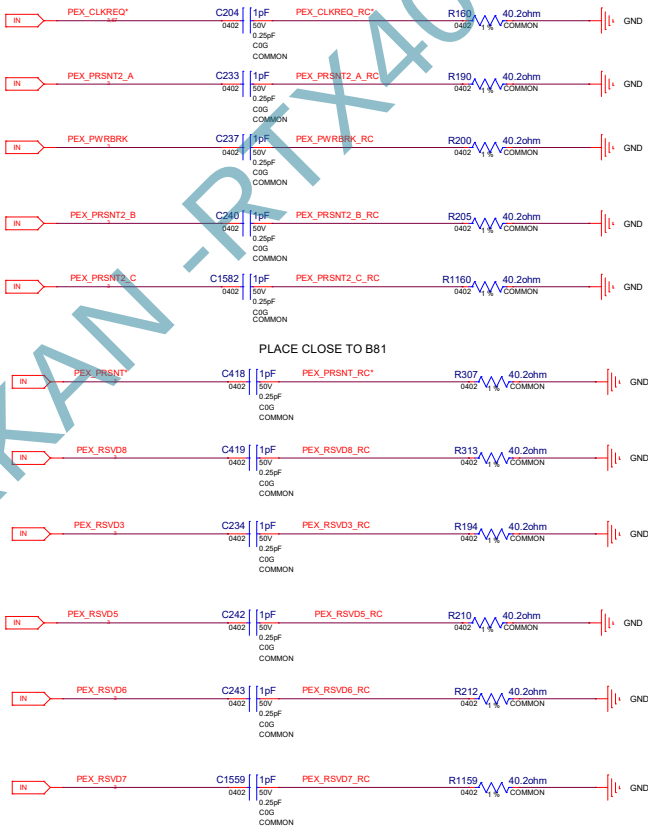
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MECH & VR THERMAL PROTECTION

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